

Safety Manager Module Installation Guide

**for use with the Triconex TRICON
Version 8 and Version 9 Systems**

SM20-600

Implementation
Safety Manager TRICON

***Safety Manager Module
Installation Guide***

**for use with the Triconex TRICON
Version 8 and Version 9 Systems**

SM20-600
9/99

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About This Publication

This publication includes both general and detailed installation information for the various major components of the Safety Manager (SM) for use with the Triconex TRICON Version 8 and Version 9 Systems. Specifically included are

- general installation instructions for the Safety Manager (SM) system,
- detailed installation instructions for installing the Safety Manager Module (SMM), and
- general installation instructions for the TRICON system.

Site planning information is contained in the *TPS System Site Planning Manual*, which is in the *System Site Planning* binder. After installation of the SM, the checkout procedure can be found in the *Safety Manager Module Service Manual*, which is in this binder.

This publication supports software Release 610.

This publication supports **TotalPlant** Solution (TPS) Network Release 610 and CE Compliant hardware. TPS is the evolution of TDC 3000^X.

All references in this manual to “Safety Manager” or “Safety Manager Module” pertain only for use with the Triconex TRICON Version 8 and Version 9 Systems.

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Acronyms

AI	Analog Input
AM	Application Module
APM	Advanced Process Manager
AO	Analog Output
DC	Digital Composite
DI	Digital Input
DO	Digital Output
HM	History Module
HPM	High Performance Process Manager
LCN	Local Control Network
NIM	Network Interface Module
PED	Point Entity Data
PI	Personality
PLC	Programmable Logic Controller
PM	Process Manager
PU	Processing Unit
RLL	Relay Ladder Logic
SM	Safety Manager
SMM	Safety Manager Module
SOE	Sequence of Events
TPS	TotalPlant Solution
UCN	Universal Control Network
US	Universal Station

References

Publication Title	Publication Number	Binder Title	Binder Number
<i>Safety Manager Module Control Functions</i>	SM09-600	Implementation Safety Manager TRICON	TPS 3074
<i>Safety Manager Module Parameter Reference Dictionary</i>	SM09-650	Implementation Safety Manager TRICON	TPS3074
<i>Safety Manager Module Configuration Forms</i>	SM88-600	Implementation Safety Manager TRICON	TPS 3074
<i>TPS System Site Planning Manual</i>	SW02-550	System Site Planning-1	TPS 3020-1
<i>Safety Manager Module Service Manual</i>	SM13-650	Implementation Safety Manager TRICON	TPS 3074

For TRICON Documentation:

Document Title	Binder Title
<i>TRICON Planning & Installation Guide</i>	TRICON Planning and Installation Guide/ User's Manual for Field Terminations
<i>TRISTATION Multi-System Workstation User's Manual</i>	TRISTATION MSW User's Manual
<i>Safety Manager Module User's Guide</i>	Manuals for Communication Products
<i>User's Manual for Field Terminations</i>	TRICON Planning and Installation Guide/ User's Manual for Field Terminations
<i>Sequence of Events User's Manual</i>	Manuals for Communication Products

Section 1 – Safety Manager System Installation

1.1 Section Overview

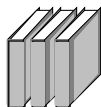
About this section

This section provides general installation information for the Safety Manager (SM) system.

Tools and test equipment

The following equipment is necessary for installation of the Safety Manager:

- Mobile lifter/lift truck—refer to Subsection 1.4—*Unpacking and Placement*.
 - Carrier band MAP tester—refer to Subsection 2.2—*Installation Considerations*.
 - Hand tools and multimeter, as in a typical electrician's tool kit.
 - Torque wrench to work in 0 to 70 cm/kg (0 to 60 inch-pound with one inch-pound resolution) range for assembly of UCN cable to their connectors.
 - Set of SAE (inch) "crow's foot" wrenches for the above torque wrench.
 - ESD kit, Honeywell part number 30185-H.
-



REFERENCE—For detailed information regarding

- site planning and installation, refer to SW02-550 *TPS System Site Planning Manual* in binder TPS 3020-1 *System Site Planning-1*.
 - post-installation checkout procedures, refer to SM13-600 *Safety Manager Module Service* in this binder.
-

1.2 Equipment Warehousing

Factory packing

For shipment, the Safety Manager is packed with desiccant packages, wrapped in a moisture barrier, placed on skids, and optionally enclosed for shipment as required by the user.

Transit/warehousing specifications

The Safety Manager can be shipped by a variety of means, but any trucking should be done by enclosed air-ride electronic vans. The equipment is usually trucked on the final leg of the journey to the user warehouse or site. There it is stored indoors until needed for installation. During the whole process, the environment must be monitored and corrected if the following equipment transit/storage ratings are exceeded:

- Temperature Range: -55° to 85°C (-67° to 158°F)
- Humidity Range: 5 to 95% RH noncondensing

ATTENTION In order to minimize exposure to humidity, keep the factory wrapping intact during storage and transit. If it is necessary to unseal the equipment for customs or receiving, add more desiccant and reseal the package.

1.3 Move Preparation

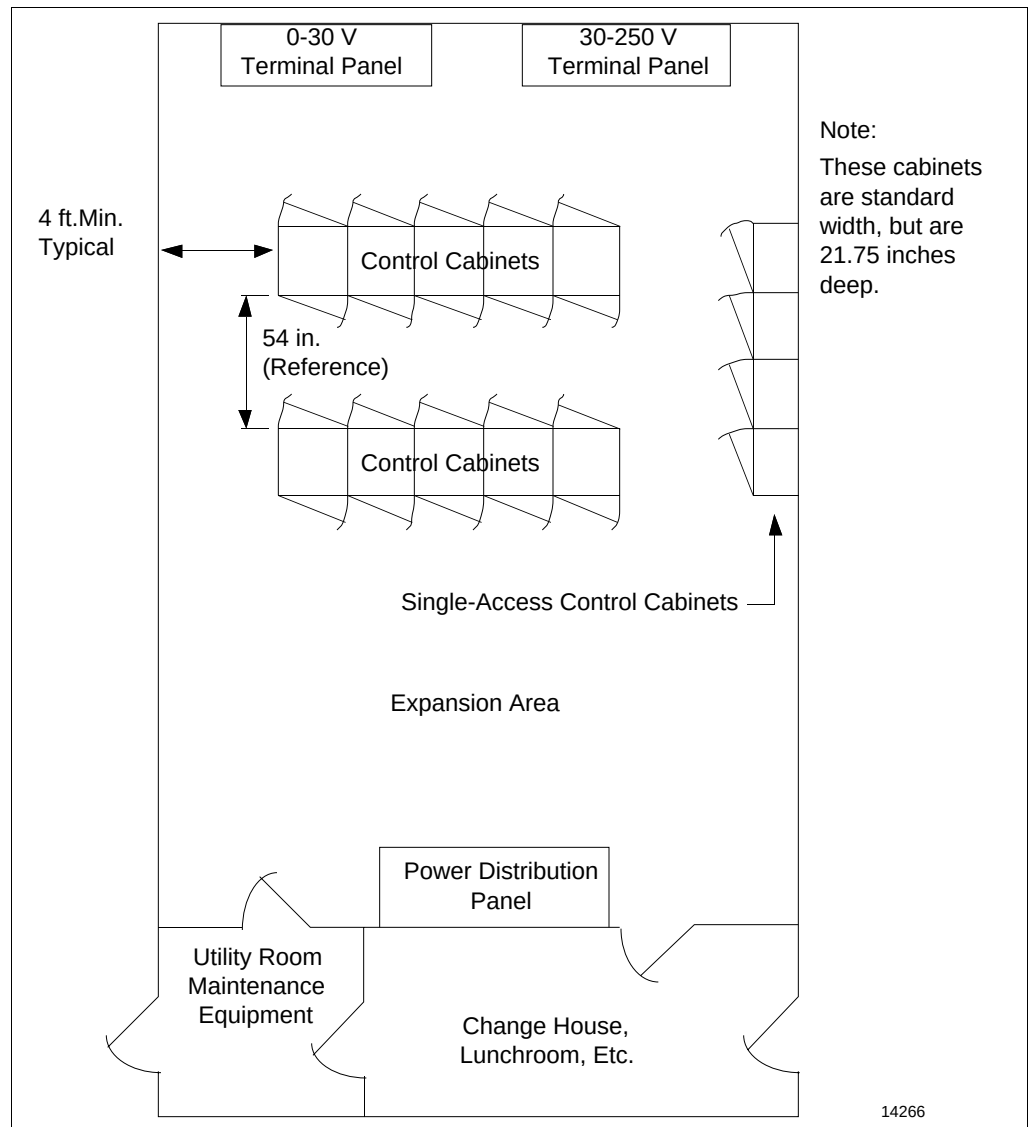
Background

Several basic points to keep in mind when planning the movement and unpacking of Safety Manager cabinets include

- a single Safety Manager cabinet is nominally 0.8 x 0.8 x 2 meters (32 x 32 x 79 inches),
- lifting eye bolts, which are removable, add about 5 cm (2 inches) to the height, and
- weight varies with options and can go as high as 385 kg (850 pounds) per cabinet.

As illustrated in Figure 1-1, a typical factory assembled complex uses up to four such cabinets bolted together.

Figure 1-1 Typical Electronics Room Layout



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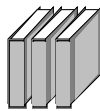
1.3 Move Preparation, Continued

Checklist

Table 1-1 provides a checklist to be used when planning the movement of Safety Manager cabinets.

Table 1-1 Move Preparation Checklist

Check	Action	Notes
1	Travel Route	Chart the equipment's route of travel regarding • loading doors—height and location, • elevators—size, capacity, and availability, • aisles—number, size, and layout, • doorways—number and size, • hallways—size, restrictions, and bends, • stairs and ramps—may require some type of lifting, and • rigging of equipment—if needed.
2	Space	Is there enough space to hold the equipment?
3	Floor Loading	Will the floor in all areas be strong enough?
4	Power Distribution	Check for proper transformer sizes, distribution panels, and sufficient outlets for support, test, and maintenance equipment.
5	Grounding	Be sure that proper grounds are available.
6	Conduits and Cable Trays	Make sure that conduits and cable trays are large enough and the proper type to accommodate all of the required cables.
7	Environment	Are lighting and other environmental requirements adequate?
8	Security	Provide security clearances for the installation crew if it is not composed of regular employees.
9	Responsibilities	Are all responsibilities clearly defined and understood?
10	Communications	At minimum, three conveniently located telephones are needed for • Honeywell Technical Assistance Center (TAC) data terminal, • Honeywell TAC voice, and • Operator Emergency.



For additional information regarding actions described in the Move Preparation Checklist, refer to SW02-550—*TPS System Site Planning Manual* in the binder TPS 3020-1 *System Site Planning-1*.

Continued on next page

1.3 Move Preparation, Continued

Preparation

Table 1-2 provides a checklist to be used when preparing the electronics room for the Safety Manager cabinets.

Table 1-2 Move Preparation Checklist

Check	Action	Notes
1	Static Electricity	Reduce the ESD borne failures by - stripping the floors of any wax to prevent static charge buildup, - using anti-static spray on floors, furniture, and fabric in the work area, and - placing a sign on or near the equipment that states circuit boards are not to be handled without wearing a ground strap.
2	Tack Rugs	Placed at all entrances, tack rugs catch dust, grit, and abrasives from other areas.
3	Fire Prevention	Consult with the local fire prevention authority to select fire extinguishers suitable for electrical fires.
4	Room Layout	Prepare an electronics room layout diagram for cabinet placement. Refer to Figure 1-1.
5	Cabinet Bolt Down	If this Safety Manager cabinet is to be bolted down, pre-drill holes in the floor. ATTENTION Refer to Figures 1-2 through 1-5 under <i>Cabinet Entry</i> in Subsection 1.3— <i>Move Preparation</i> for placement and dimensions of the bolt down holes.

Continued on next page

1.3 Move Preparation, Continued

Cabinet entry

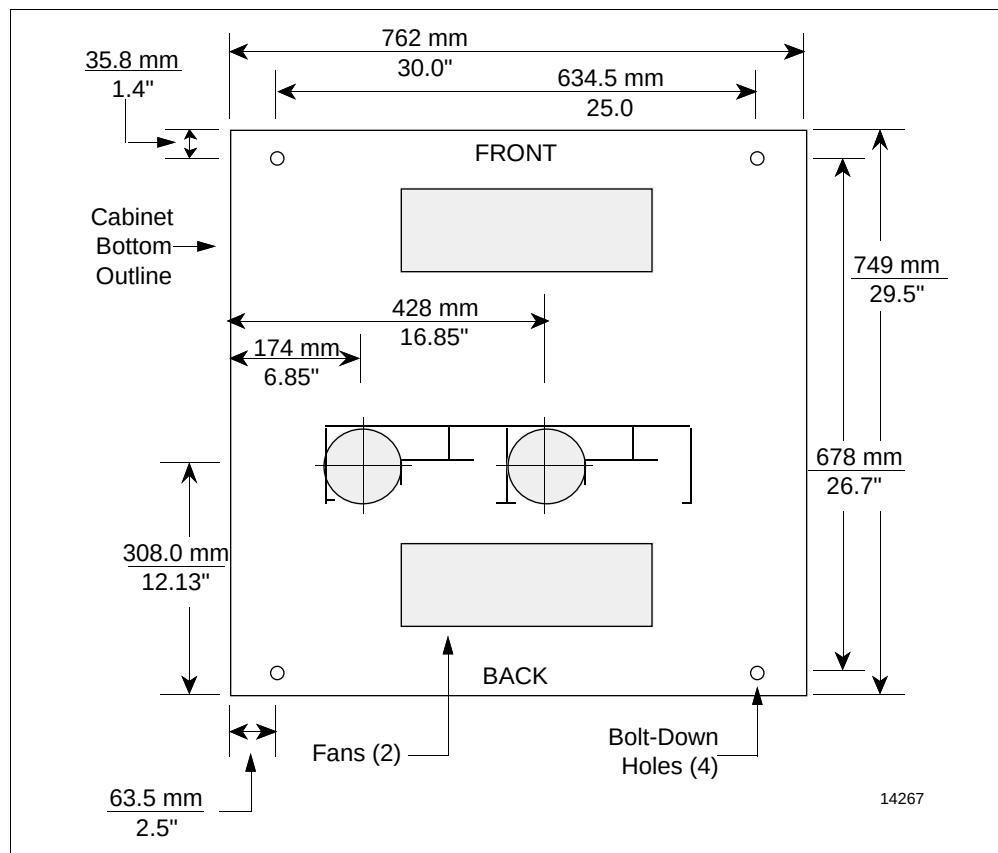
The process signal cables can enter the Safety Manager cabinet through the top or bottom of the cabinet. Figures 1-2 through 1-5 illustrate the placement and dimensions for process wiring.

The Safety Manager cabinet is special, because for bottom entry it has sliding floor plates that can be adjusted to provide entry slots. For top entry, use a hydraulic hole punch to minimize the generation of metal filings. For top or bottom entry, the cables should be clamped firmly to the cabinet. The clamping should be able to handle about a 45 kg (100 pound) pull. This prevents damage inside the cabinet when pulling on cables outside the cabinet.

The Safety Manager cabinet includes a cable clamp rail at the bottom. The rail can be remounted to the top if required.

Figure 1-2 illustrates a Safety Manager cabinet bottom configured for two 254 mm (10.0") cable trays. Also depicted are the cabinet's bolt-down holes and dimensions.

Figure 1-2 SM Cabinet with Two 254 mm (10.0") Trays



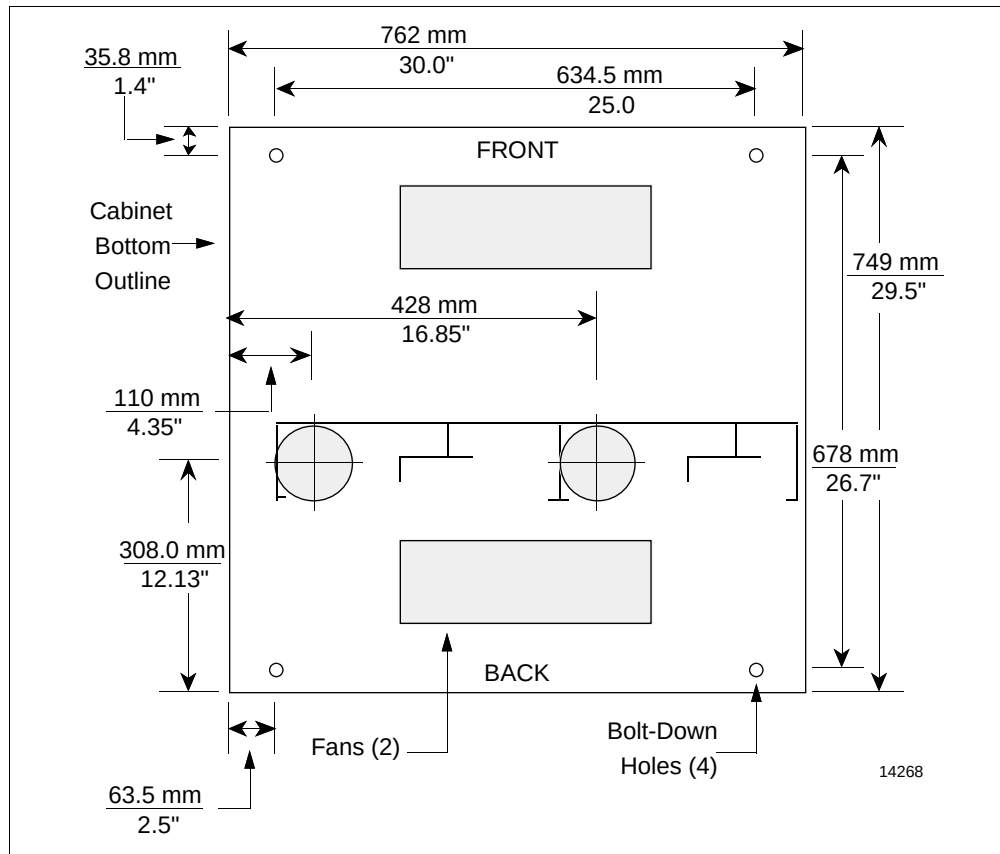
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1.3 Move Preparation, Continued

Cabinet entry,
continued

Figure 1-3 illustrates a Safety Manager cabinet bottom configured for two 381 mm (15.0") cable trays. Also depicted are the cabinet's bolt-down holes and dimensions.

Figure 1-3 SM Cabinet with Two 381 mm (15.0") Trays



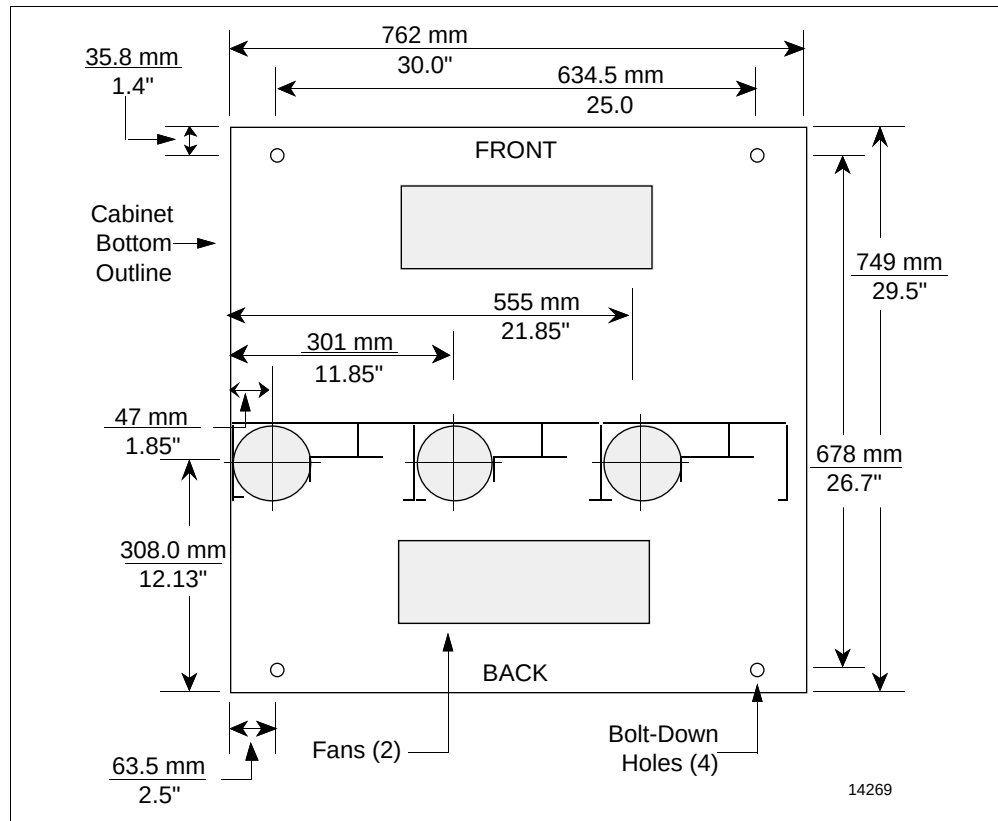
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1.3 Move Preparation, Continued

Cabinet entry,
continued

Figure 1-4 illustrates a Safety Manager cabinet bottom configured for three 254 mm (10.0") cable trays. Also depicted are the cabinet's bolt-down holes and dimensions.

Figure 1-4 SM Cabinet with Three 254 mm (10.0") Trays



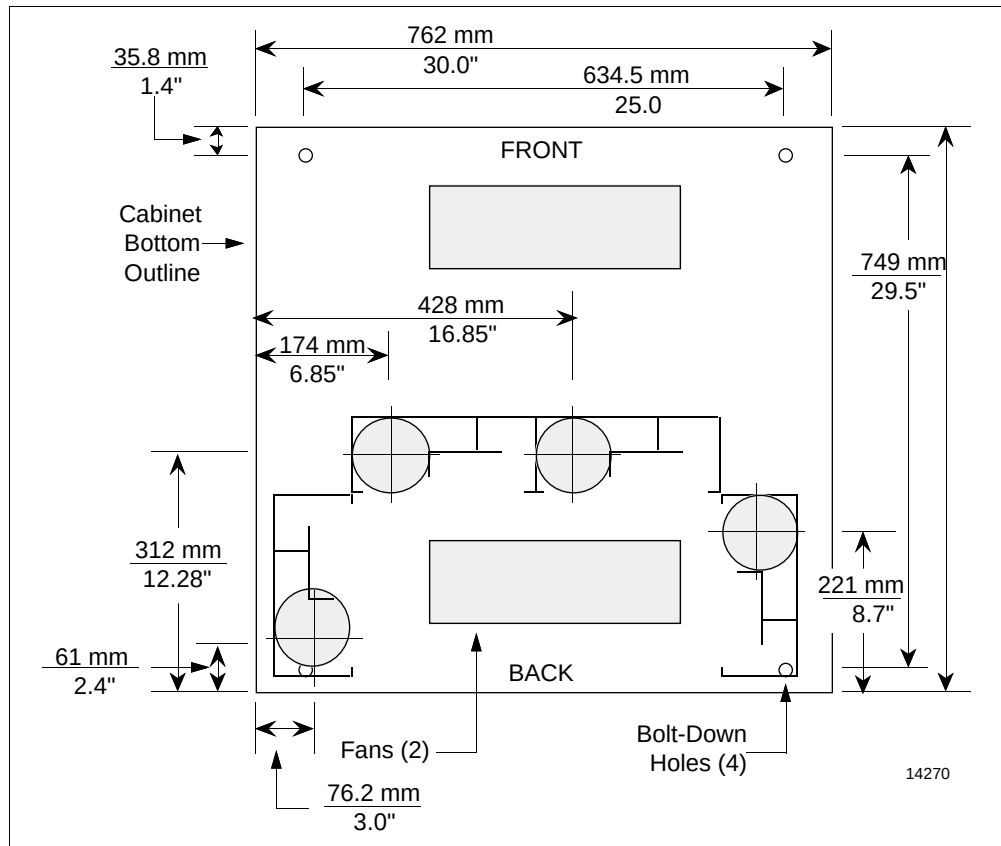
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1.3 Move Preparation, Continued

Cabinet entry,
continued

Figure 1-5 illustrates a Safety Manager cabinet bottom configured for four 254 mm (10.0") cable trays. Also depicted are the cabinet's bolt-down holes and dimensions.

Figure 1-5 SM Cabinet with Four 254 mm (10.0") Trays



1.4 Unpacking and Placement

Unpacking

Unpack the equipment, check the shipment against the invoice, and notify Honeywell representative of any shortages immediately. Table 1-3 lists important considerations when unpacking the equipment.

Table 1-3 Important Unpacking Considerations

IF ...	THEN ...
the Safety Manager system is comprised of individual card files	the component parts are packaged in labeled boxes.
a Product Registration Label containing the Model Number and Serial Number of the component is affixed to the shipping carton when received	remove and return the label to Honeywell at the noted address to ensure follow-up service and support.

Moving

The Safety Manager equipment can be moved in any one of four methods:

- fork lift,
- mobile lifter,
- roller, or
- crane.

Special consideration for each of these methods is provided in the following information.

CAUTION

When moving the Safety Manager equipment

- use lifters with a capacity of 1100 kg (2500 lbs) or greater, and
- do not tilt the cabinet more than 45° front or back to avoid damaging the doors; they are not recessed.

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1.4 Unpacking and Placement, Continued

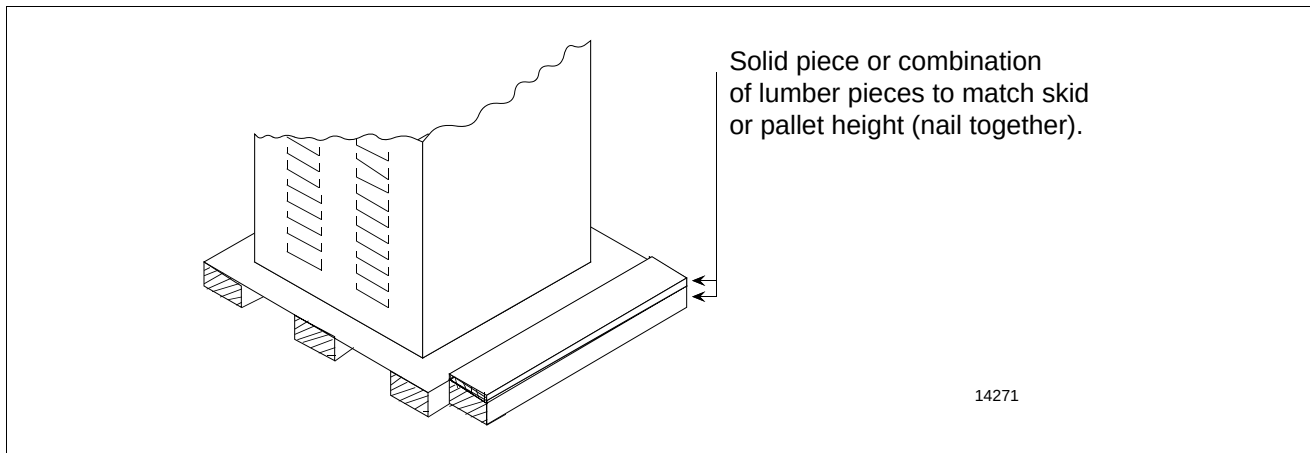
Fork lift

The Safety Manager, on the original shipping skid, is moved by fork lift to the final location. Table 1-4 lists special considerations regarding the use of a fork lift to move the equipment.

Table 1-4 Fork Lift Considerations

Number	Consideration
1	Move the equipment to the electronics room by fork lift; remove the fork lift.
2	Open the cabinet doors, locate the skid bolts, and remove them.
3	Stack lumber against the pallet to a matching height, as illustrated in Figure 1-6.
4	Manually move the equipment onto the lumber by 10 cm (4 inches).
5	Tip the cabinet weight onto the lumber and pull out the skid.
6	Tip the cabinet off the lumber and onto the floor; remove the lumber.
7	Manually move the cabinet to the exact final position.

Figure 1-6 Removal of Equipment from Shipping Skid



Continued on next page

1.4 Unpacking and Placement, Continued

Mobile lifter

As illustrated in Figure 1-7, the Safety Manager is removed from the shipping skid and placed on a transport skid made from angle iron. It is then moved with a mobile lifter. Table 1-5 lists special considerations regarding the use of a mobile lifter to move the equipment.

Table 1-5 Mobile Lifter Considerations

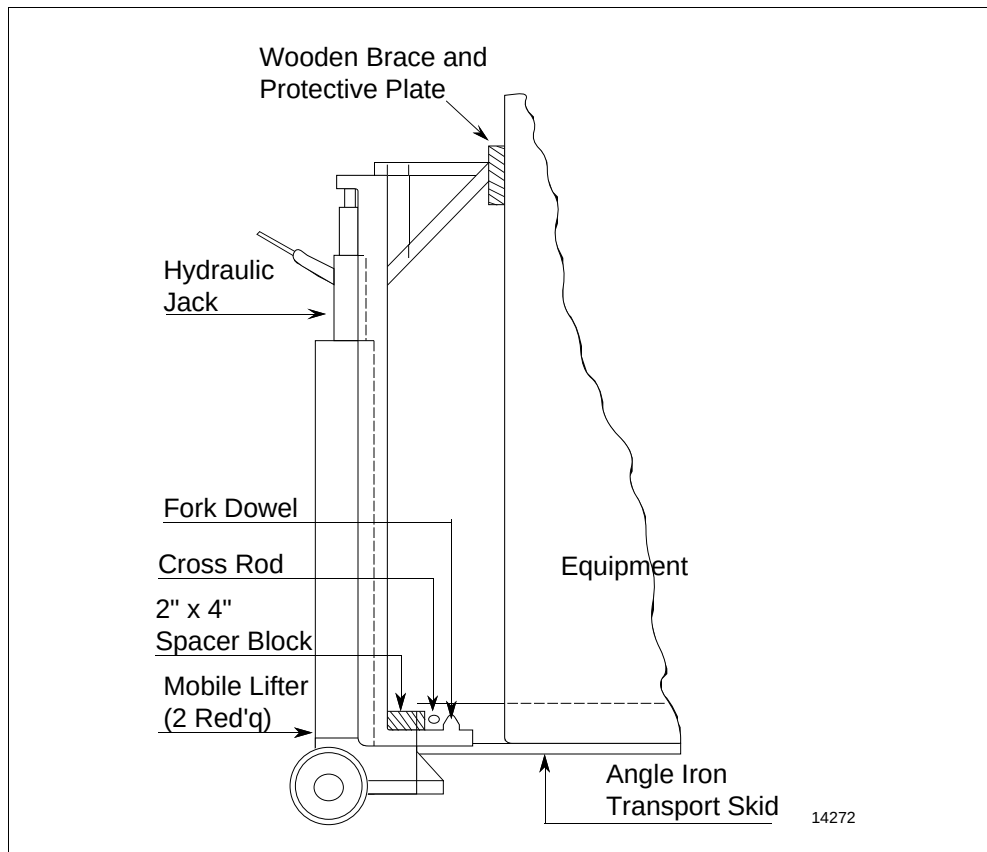
Number	Consideration
1	Remove any steel bands, turnbuckles, and cables securing the equipment to the shipping skid.
2	Open the cabinet doors and locate the bolts holding the cabinets to the shipping skid; remove them and close the doors.
3	Tilt the equipment slightly and slide one of the angle irons used to make up a transport skid underneath the cabinet as in Figure 1-7.
4	Repeat the above step for the other angle.
5	Insert a strip of heavy cardboard between each angle iron and the equipment finish.
6	Tighten the angle irons against the cabinet by tapping slightly.
7	Insert and tighten two 12 mm (1/2-inch) minimum, threaded cross rods as in Figures 1-7 and 1-8.
8	Spread the lifter forks to 0.8 m (32-inch) centers and put them under each cross rod. Place the fork's dowels between the cross rods and the cabinet, just touching the cross rods.
9	Place wooden spacer blocks along the forks, long enough to press the cross rods against the fork dowels.
10	Position a protective plate using braces as in Figure 1-7. Be sure that the plate does not press against handles or other cabinet hardware.
11	Raise both lifter forks evenly until the equipment clears the shipping skid; pull it out of the way.
12	Wheel the equipment to the final position in the electronics room; lower it to the floor.
13	Pull out the mobile lifters. Dismantle and remove the angle iron transport skid.
14	Manually place the cabinet in the exact final position.

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1.4 Unpacking and Placement, Continued

Mobile lifters,
continued

Figure 1-7 Removal of Equipment from Shipping Skid—Part 2

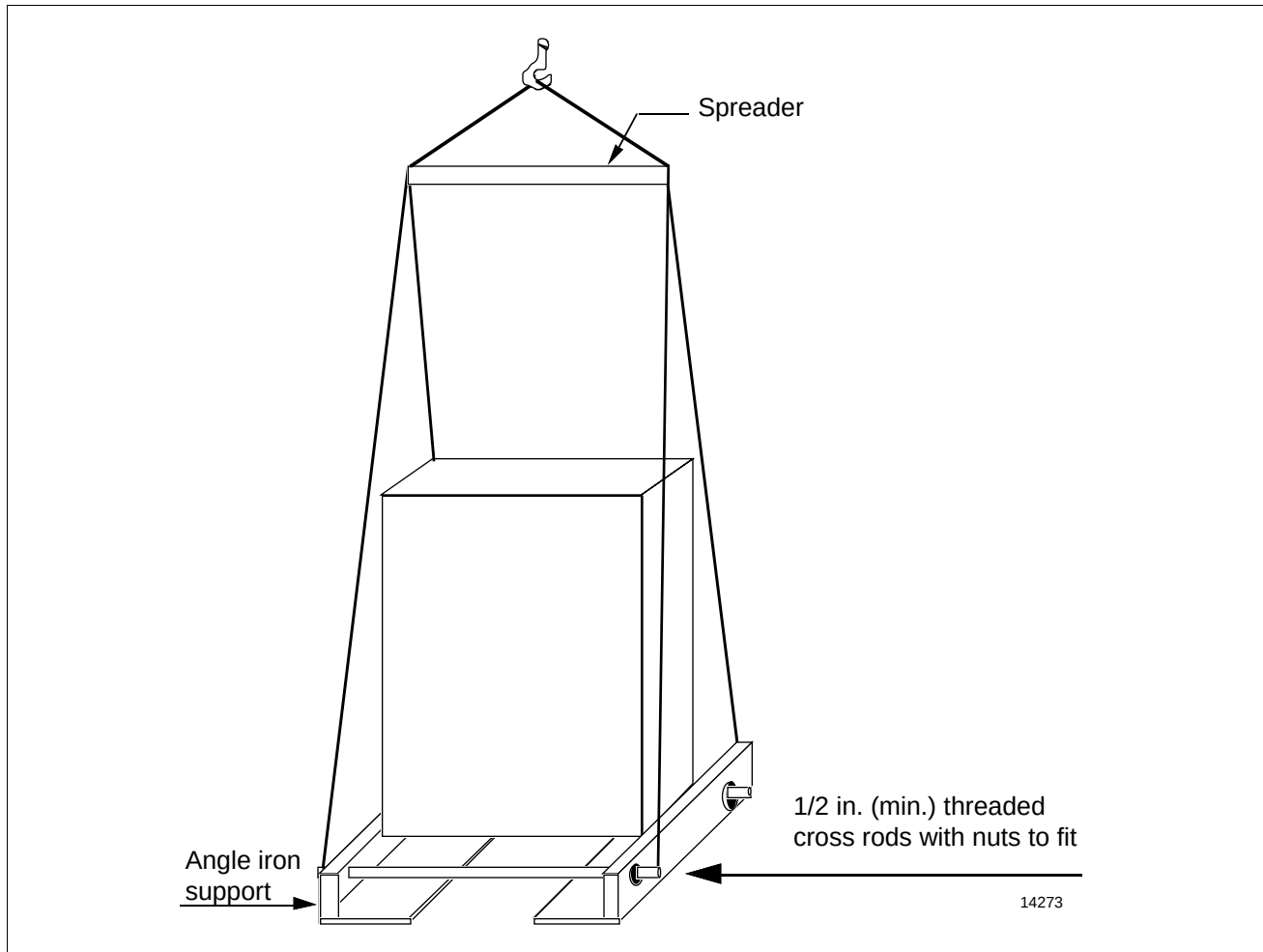


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1.4 Unpacking and Placement, Continued

Mobile lifters, continued

Figure 1-8 Lifting Equipment by Mobile Lifter or Crane



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1.4 Unpacking and Placement, Continued

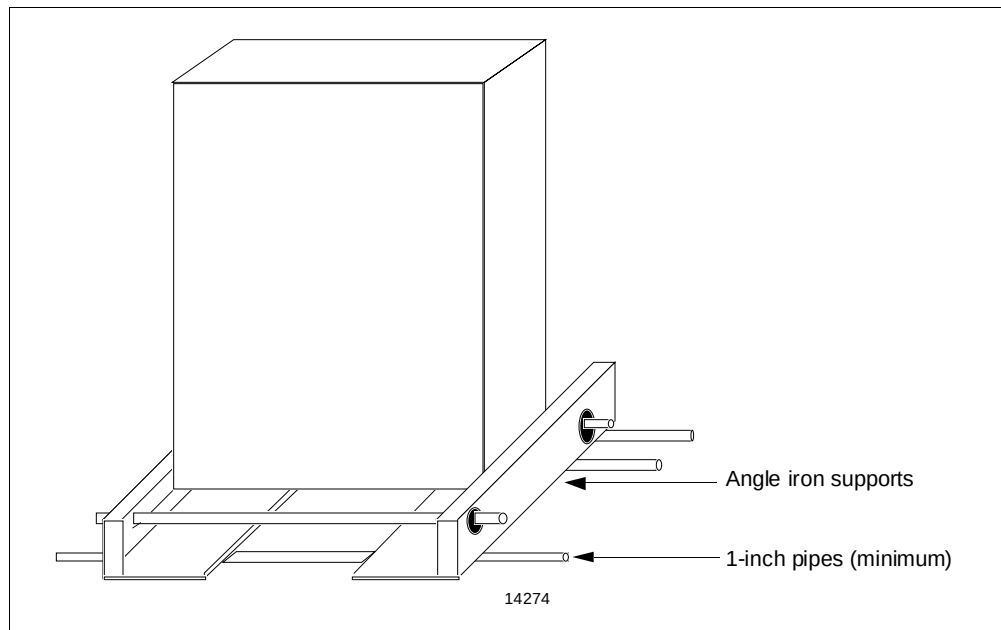
Roller method

As illustrated in Figure 1-9, using the roller method the Safety Manager is moved to its final destination by placing pipe rollers under a transport skid made of angle iron. Table 1-6 lists special considerations regarding the use of the roller method to move the equipment.

Table 1-6 Roller Method Considerations

Number	Consideration
1	Use one of the previous methods to move the cabinet to a location where rollers are to be used.
2	Install the angle iron transport skid as in the mobile lifter method.
3	Place 12 mm (1 inch) pipes under the transport skid as in Figure 1-9. • The load must be on the angle irons, not the cabinet.
4	Roll the cabinet to the final location. • At least three pipes must be under the transport skid at all times.
5	When the cabinet is near the final location, tilt it and pull out the pipes, the middle one last.
6	Dismantle the transport skid and position the cabinet manually to the exact final location.

Figure 1-9 Lifting Equipment by Mobile Lifter or Crane—Part 2



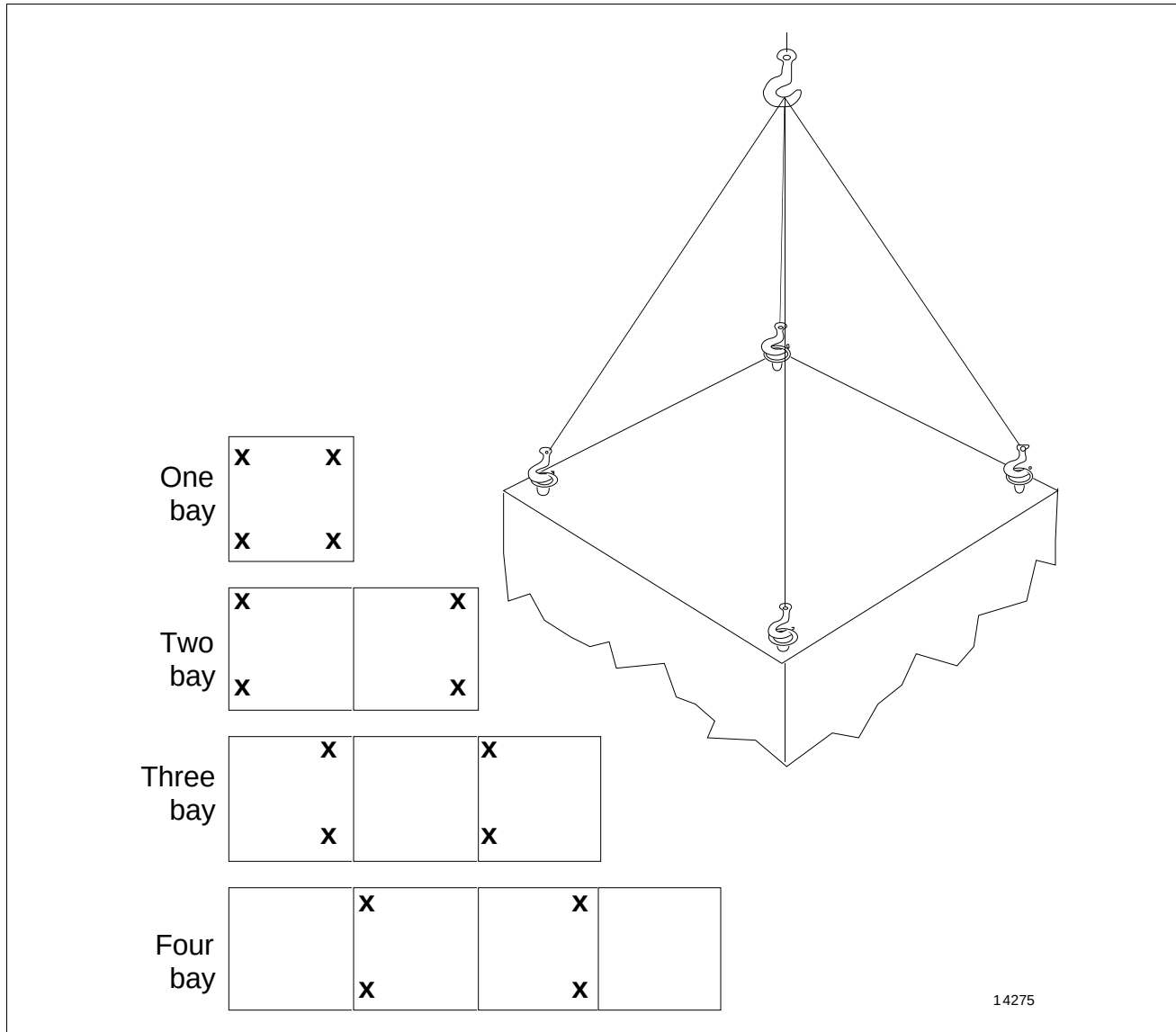
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1.4 Unpacking and Placement, Continued

Crane method

In this method the cabinet is lifted into place with a crane. Use either an angle iron transport skid (Figure 1-9) or the lifting eyebolts installed on each cabinet (Figure 1-10).

Figure 1-10 Eye Bolt Crane Method



Leveling and bolt down

Safety Manager cabinets do not include levelers; it may be necessary to hand-fit shims underneath the cabinet in order to make the cabinet sit solidly.

The SM cabinet has six holes in the bottom for bolting it to a floor. A partially filled cabinet should be bolted down. Cabinets that are bolted together are not easily moved and do not ordinarily require bolting down. Refer to Figures 1-2 through 1-5 for bolt-down dimensions.

Section 2 – System Component Installation

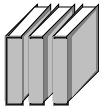
2.1 Section Overview

About this section

This section provides general installation information for various components of the Safety Manager (SM) system.

ATTENTION

This section describes in detail, only those installation procedures and considerations specific to the Safety Manager Module and the **TotalPlant** Solution (TPS) system installation procedures and considerations specific to the TRICON components can be located in the Triconex documentation.



REFERENCE—For detailed information regarding

- site planning and installation, refer to SW02-550 *TPS System Site Planning Manual* in binder TPS 3020-1 *System Site Planning-1*.
 - post-installation checkout procedures, refer to SM13-600 *Safety Manager Service Manual* in this binder.
-

2.2 Installation Considerations

Introduction

The Safety Manager system may encompass a single cabinet, a multi-cabinet complex, individual card files, and supporting power supply modules, or a combination of such hardware.

If your SM system involves only a cabinet or cabinets, then power, ground, and cabling are brought to the system and connected. The system is configured and installation is complete.

Should your SM system not include a cabinet or cabinets, the individual chassis must be mounted to a structure and power and ground supplied by a choice of methods.

ATTENTION The first time an SMM is inserted into the Triconex chassis and powered up, it will cause a chassis alarm to be generated. The SMM enters the “Alive State” waiting for the SMPI software to be downloaded. At this point, the SMM is unable to respond to a command by TSX to go active, therefore causing the alarm. The SMM will not go active until a personality is loaded into the SMM from the TPS system.

Installation checklist

When installing an SM system, consider the following installation checklist:

- Carefully review all documentation available for the equipment, paying close attention to cautionary notes.
 - Install the Safety Manager system equipment in accordance with the current edition of the National Electric Code ANSI/NFPA 70.
 - Use technically qualified personnel to service the Safety Manager system components.
-

2.3 Cabinet Installation

Introduction

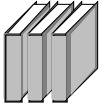
After placement, the cabinet or cabinet complex requires the attachment of power and ground, cabling installation, and if necessary, configuration of certain card file modules.

ATTENTION Refer to Section 1—*Safety Manager System Installation* for information regarding the warehousing, movement, unpacking, and placement of the Safety Manager cabinets.

2.4 Card File Installation

Introduction

The three TRICON chassis associated with the Safety Manager are identical in dimensions and necessary clearances. Their total weight is dependent upon the modules installed.



Refer to Chapter 4—*Installation and Checkout*, Subsection—*Mechanical Installation*, in the *TRICON Planning and Installation Guide* for information regarding the physical installation procedures and considerations of the TRICON chassis.

2.5 Cabling and Wiring Considerations

Introduction

When installing a Safety Manager System, it is important to consider:

- conduit and tray shielding and spacing, and
- cable and wire classifications.

Shielding

Conduit or cable trays should be used to route the network trunk line within the plant for short distances or for several miles. The same conduit should not carry both data and power wiring because of problems created by electromagnetic interference (EMI). The choice of materials and configuration for the cable conduit can determine the degree of protection from this interference.

Table 2-1, showing raceway shielding, is taken from IEEE Standard 518 for the installation of electrical equipment. It compares different types of conduits and raceways for magnetic field attenuation at 60 Hz and for electrostatic attenuation at 100 kHz.

Table 2-1 Raceway Shielding

Raceway Type	Thickness mm (in.)	60 Hz Magnetic Field Attenuation		100 Hz Magnetic Field Attenuation	
		Ratio	db	Ratio	db
Free air		1:1	0	1:1	0
2 inch aluminum conduit	3.91 (0.154)	1.5:1	3.3	2,150:1	66.5
#16 gauge aluminum tray	1.52 (0.060)	1.6:1	4.1	15,500:1	83.9
#16 gauge aluminum tray	1.52 (0.060)	3:1	9.4	20,000:1	86.0
#16 gauge galvan. iron tray	1.52 (0.060)	3.2:1	10.0	22,000:1	86.8
2 inch IPS copper pipe	3.96 (0.156)	3.3:1	10.2	10,750:1	80.6
#16 gauge aluminum tray	1.52 (0.060)	4.2:1	11.5	29,000:1	89.3
#14 gauge aluminum tray	1.90 (0.075)	6:1	15.5	23,750:1	87.5
2 inch metallic tubing	1.65 (0.065)	6.7:1	16.5	3,350:1	70.5
2 inch rigid galvanized conduit	3.91 (0.154)	40:1	32.0	8,850:1	78.9

Continued on next page

2.5 Cabling and Wiring Considerations, Continued

Spacing

It is recommended that noise creating sources (those with time varying voltage or time varying current) be separated from data signal cable. Cables with similar levels of noise susceptibility should be grouped together, and those with similar levels of noise generation should be separately grouped in trays and conduits.

IEEE 518 defines four classes of wiring that differ in signal level and noise susceptibility as provided in Table 2-2.

Table 2-2 IEEE 518 Wiring Classification Definitions

Class	Definition
Data 1	High noise susceptibility. Analog signals of less than 50 V and digital signals of less than 15 V.
Data 2	Medium susceptibility. Analog signals greater than 50 V, regulating signals of 50 V with currents less than 20 A; ac feeders less than 20 A.
Data 3	Low susceptibility. Switching signals greater than 50 V, analog signals of 50 V with currents less than 20 A; ac feeders less than 20 A.
Data 4	Power ac and dc buses of 0-1000 V with currents of 20-800 A.

ATTENTION Data 3 and Data 4 classes have one subclass each for applications that require special cable and spacing. Tables 2-3, 2-4, and 2-5 in this section list the subclass for Data 3—listed as Data 3A; the subclass for Data 4 is listed as Data 4A. Examples of Data 3A and Data 4A subclass cables include

- signals from communicating fields and line resistors,
- signals from shunts to regulators, and
- power greater than 1000 volts or greater than 800 amperes, or both.

Tables 2-3, 2-4, and 2-5 list the recommended minimum distances between trays, trays and conduits, and conduits, respectively. The dimensions provided in the tables are between the top of one tray/conduit and the bottom of the tray/conduit above, or between sides of adjacent trays/conduits.

Trays containing Data 1 or Data 2 cables should have solid bottoms and covers to complete shielding. Ventilation slots or louvers may be used only in trays containing Data 3 cables.

Continued on next page

2.5 Cabling and Wiring Considerations, Continued

Spacing, continued

Table 2-3 Tray Spacing

Cable Class	Data 1 mm (in.)	Data 2 mm (in.)	Data 3 mm (in.)	Data 3A mm (in.)	Data 4 mm (in.)	Data 4A mm (in.)
Data 1		<i>Note 2</i>	152.4 (6)	152.4 (6)	660.4 (26)	660.4 (26)
Data 2	<i>Note 2</i>		152.4 (6)	152.4 (6)	457.2 (18)	660.4 (26)
Data 3	152.4 (6)	152.4 (6)		0.0 (0)	<i>Note 1</i>	304.8 (12)
Data 3A	152.4 (6)	152.4 (6)	0.0 (0)		203.2 (8)	457.2 (18)
Data 4	660.4 (26)	457.2 (18)	<i>Note 1</i>	203.2 (8)		0.0 (0)
Data 4A	660.4 (26)	660.4 (26)	304.8 (12)	457.2 (18)	0.0 (0)	

ATTENTION

Note the following considerations:

- Note 1
 - Data 3 and Data 4 cables may be run in a common tray but should be separated by a barrier. This barrier does not have to be grounded.
- Note 2
 - Data 1 and Data 2 cables may be in a common tray, provided they are separated by a grounded steel barrier.

Table 2-4 Tray/Conduit Spacing

Cable Class	Data 1 mm (in.)	Data 2 mm (in.)	Data 3 mm (in.)	Data 3A mm (in.)	Data 4 mm (in.)	Data 4A mm (in.)
Data 1		25.4 (1)	101.6 (4)	101.6 (4)	457.2 (18)	457.2 (18)
Data 2	25.4 (1)		101.6 (4)	101.6 (4)	304.8 (12)	457.2 (18)
Data 3	101.6 (4)	101.6 (4)		0.0 (0)	0.0 (0)	203.2 (8)
Data 3A	101.6 (4)	101.6 (4)	0.0 (0)		152.4 (6)	304.8 (12)
Data 4	457.2 (18)	304.8 (12)	0.0 (0)	152.4 (6)		0.0 (0)
Data 4A	457.2 (18)	457.2 (18)	203.2 (8)	304.8 (12)	0.0 (0)	

Continued on next page

2.5 Cabling and Wiring Considerations, Continued

Spacing, continued

Table 2-5 Conduit Spacing

Cable Class	Data 1 mm (in.)	Data 2 mm (in.)	Data 3 mm (in.)	Data 3A mm (in.)	Data 4 mm (in.)	Data 4A mm (in.)
Data 1		25.4 (1)	76.2 (3)	76.2 (3)	304.8 (12)	304.8 (12)
Data 2	25.4 (1)		76.2 (3)	76.2 (3)	228.6 (9)	304.8 (12)
Data 3	76.2 (3)	76.2 (3)		0.0 (0)	0.0 (0)	152.4 (6)
Data 3A	76.2 (3)	76.2 (3)	0.0 (0)		152.4 (6)	228.6 (9)
Data 4	304.8 (12)	228.6 (9)	0.0 (0)	152.4 (6)		0.0 (0)
Data 4A	304.8 (12)	304.8 (12)	152.4 (6)	228.6 (9)	0.0 (0)	

Cable, wire, and conduit selection

Select the proper cables for each type of system connection. All Programmable Logic Controller (such as the TRICON) system wires and cables can be classified into two categories:

- High level conductors
 - used to carry high voltage and/or high current,
 - typically IEEE 518 Data 2, Data 3, and Data 4 categories, and
 - include AC I/O lines; high level DC I/O lines connected to such devices as switches, relays, or solenoids; and ac power lines.
- Low level conductors
 - used to carry low voltage and/or low current,
 - typically IEEE 518 Data 1 category, and
 - include communication cables; DC I/O lines connected to solid state devices such as proximity switches or photoelectric sensors; TTL signal lines; analog signal lines; encoder signal lines; and motion control signal lines.

Continued on next page

2.5 Cabling and Wiring Considerations, Continued

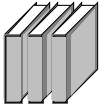
Cabling and wiring guidelines

Observe the guidelines presented in Table 2-6 when you route cables to the Safety Manager and its remote enclosures. Pay special attention to separating data cables from power cables to avoid electrical noise problems.

Table 2-6 Cabling and Wiring Guidelines

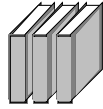
Consideration	Description
1	Maintain a 60°C (140°F), or less, ambient air temperature at the bottom of each Safety Manager system card file.
2	Do all wiring according to IEEE Standard 518-1977.
3	When you use a wire duct less than 3.5 inches deep, allow at least 1.5 inches between it and the rack assembly. For a wire duct greater than 3.5 inches deep, allow clearance of 2 inches.
4	Separate all 440 Vac (or higher) electrical supply lines as far from Safety Manager System components as possible. See Tables 2-3, 2-4, and 2-5 for the recommended minimum distances.
5	Run all 440 Vac lines in separate conduits from I/O wiring ducts.
6	Separate all ac and low level dc wiring. See Tables 2-3, 2-4, and 2-5 for the recommended minimum distances.
7	When 8 inch deep enclosures are used, door stiffeners are not permitted in front of the Safety Manager system components.
8	Shields of data communications cables must be terminated at both ends. Shields of analog signal cables must be terminated to ground only at the signal source. If a termination is not made, or is intermittent, noise immunity is lost.
9	Module thumbscrews must be tightened to ensure proper grounding for the signal shield connections.
10	Data and power cables should cross at right angles and not run in parallel for any distance.
11	Ensure that the entire communications network system is grounded to earth ground.
12	Maintain shield continuity. Ensure that shield leads are not broken.
13	Route cables around rather than through high noise areas.

2.6 Wiring and Cabling Installation



Card file to card file cabling

REFERENCE—For information regarding general wiring and cabling considerations, refer to Subsection 2.4—*Cabling and Wiring Considerations*.



Refer to the following references in your Triconex documentation.

- For considerations and specifications regarding the I/O bus cabling between chassis, refer to
 - Chapter 2—*Basic Components*, Subsection—*I/O Expansion*, in the *TRICON Planning and Installation Guide*
 - Chapter 4—*Installation and Checkout*, Subsection—*Connecting Chassis with the I/O Bus*, in the *TRICON Planning and Installation Guide*
 - Appendix D—*Pin-Outs for TRICON Connectors*, Subsection—*I/O Bus Cable for Interconnected Chassis*, in the *TRICON Planning and Installation Guide*
 - For special considerations and specifications regarding RXM chassis, refer to Chapter 6—*Using Remote Extender Chassis* in the *TRICON Planning and Installation Guide*
 - For recommended wiring methods, refer to Appendix F—*Recommended Wiring Methods*, in the *TRICON Planning and Installation Guide*
-

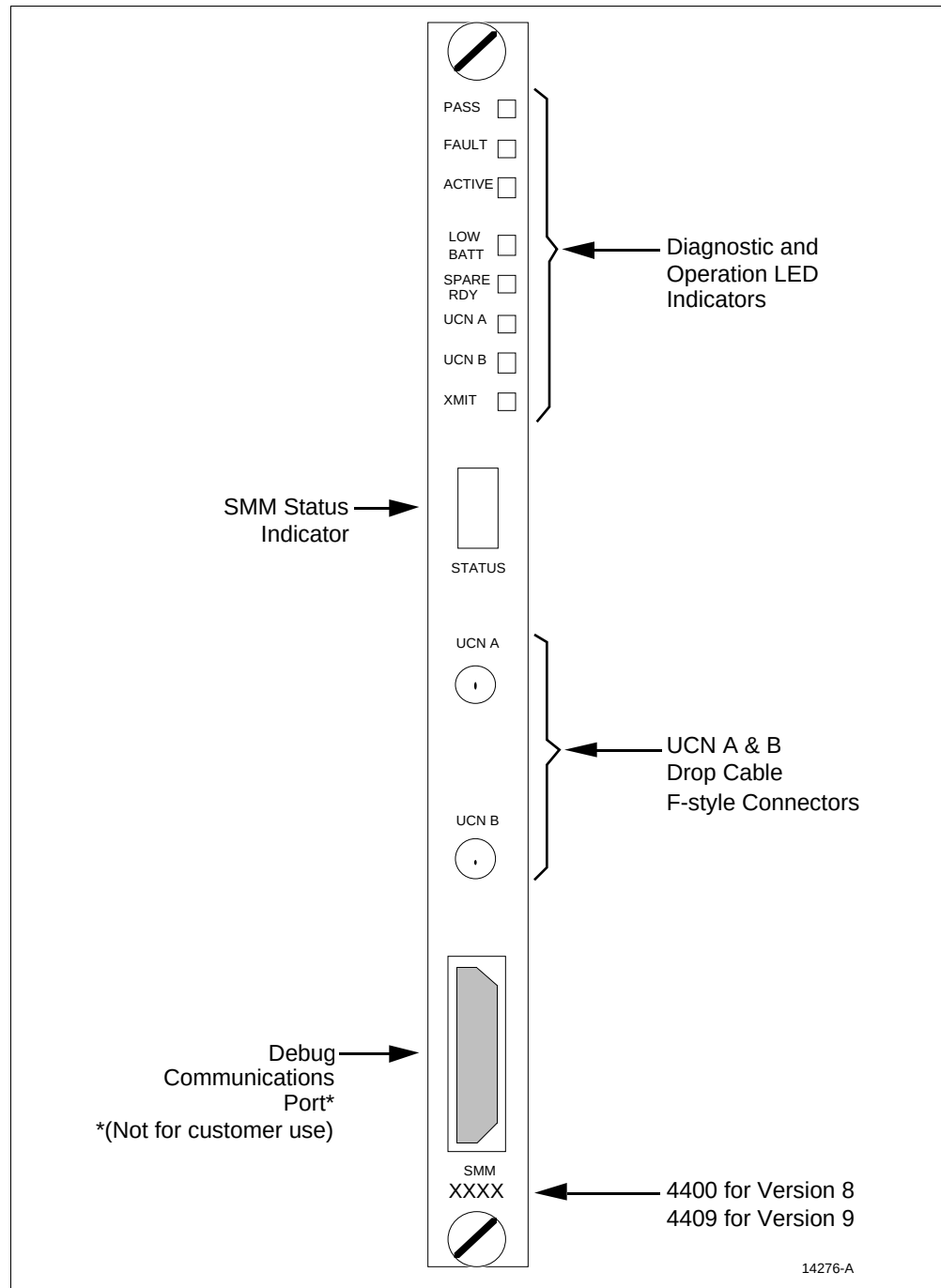
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2.6 Wiring and Cabling Installation, Continued

SMM cabling

Figure 2-1 illustrates the frontplate of the Safety Manager Module (SMM). Connections to the UCN are made via the two UCN F-style connectors in the center of the module.

Figure 2-1 SMM Frontplate



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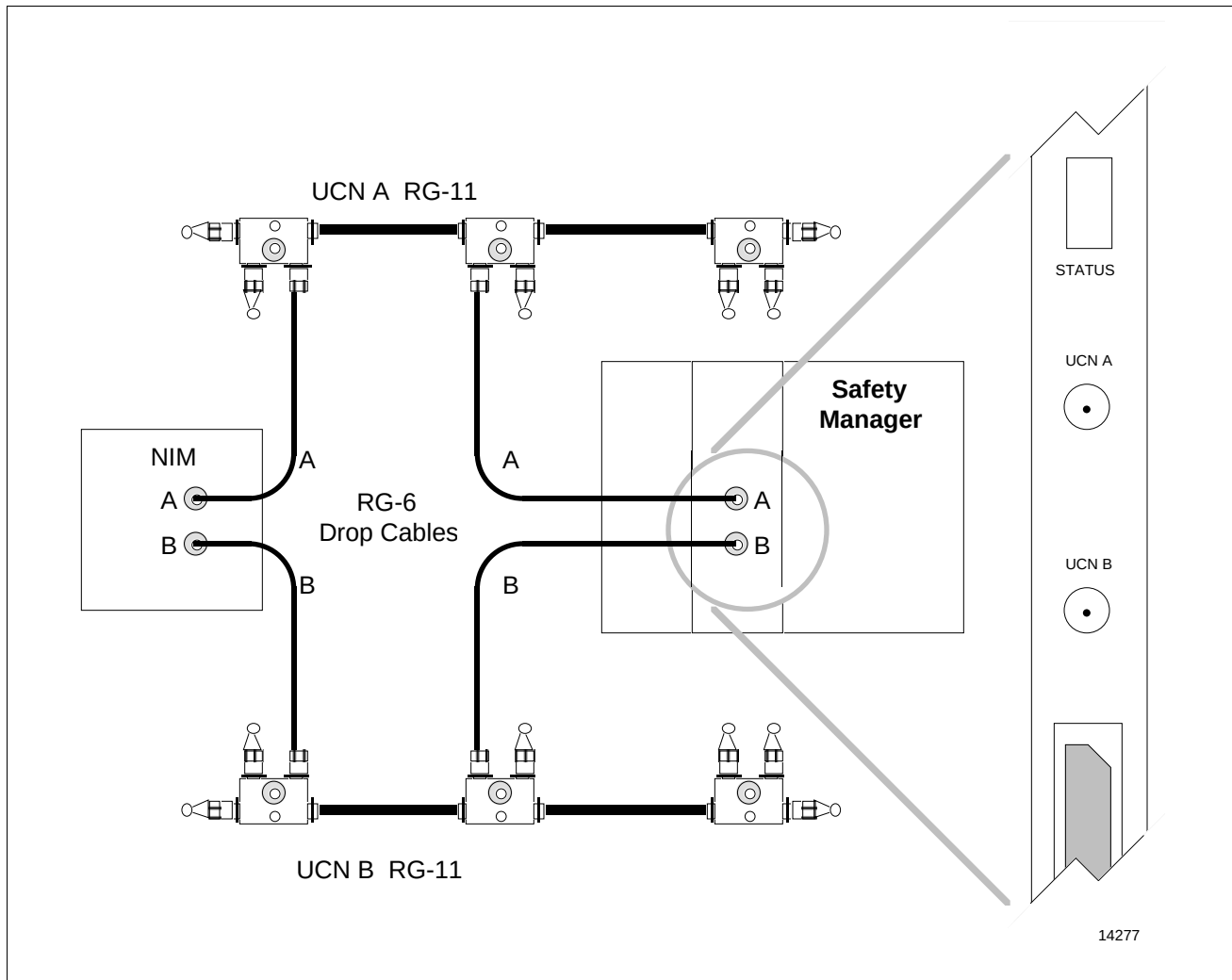
2.6 Wiring and Cabling Installation, Continued

SMM cabling, continued

Figure 2-2 illustrates UCN cabling for a non-redundant Safety Manager system.

In this case the SMM is installed in center or right-hand physical slot of a TRICON's logical slot. A single RG-6 UCN drop cable is run from the drop port on UCN A to F-style connector A of the SMM. A single RG-6 UCN drop cable is also run from the drop port on UCN B to F-style connector B of the SMM.

Figure 2-2 Non-Redundant SM UCN Cabling



CAUTION Remove UCN cable from SMM port, without breaking UCN cable continuity, or communication will be lost to other UCN devices.

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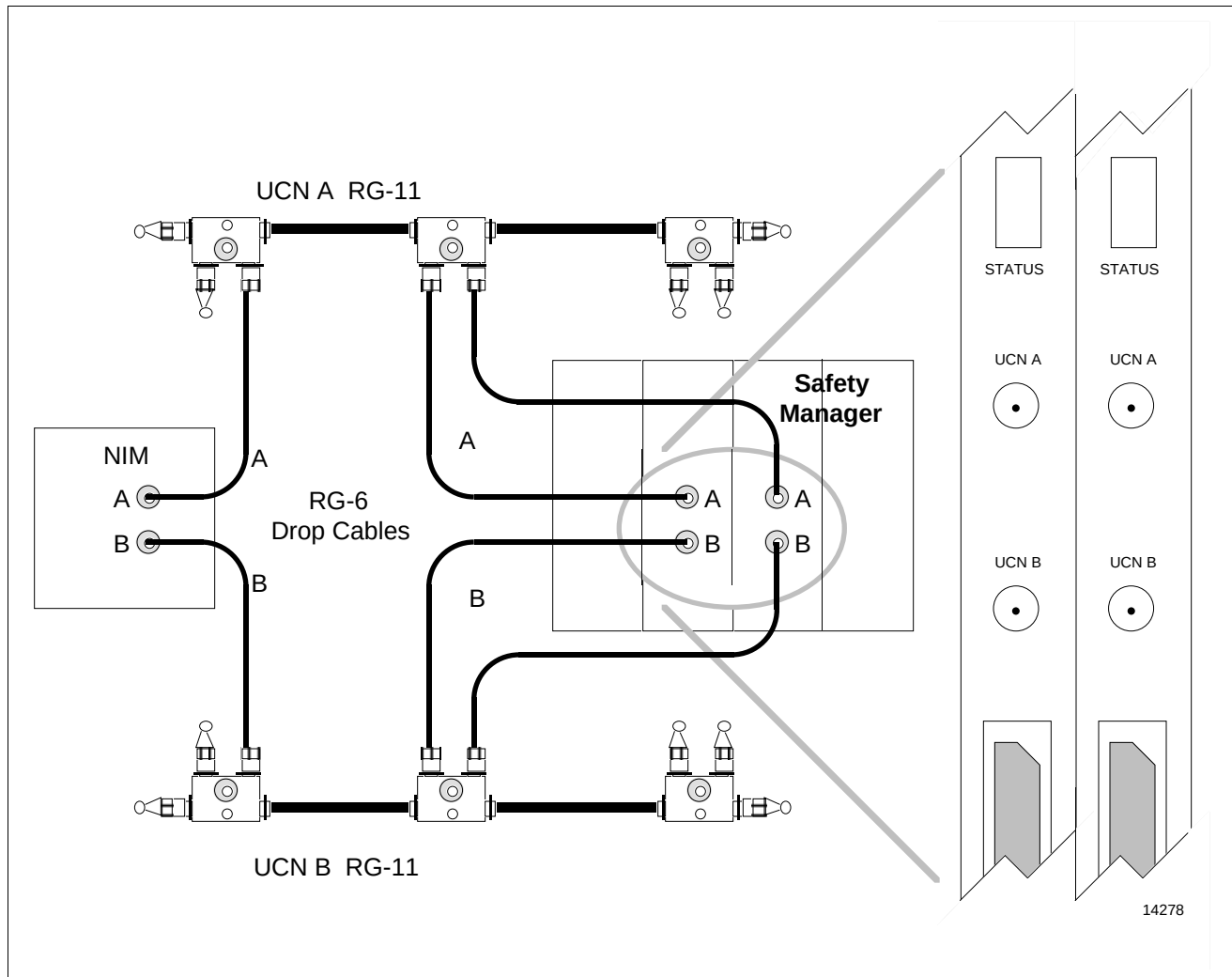
2.6 Wiring and Cabling Installation, Continued

SMM cabling, continued

Figure 2-3 illustrates the UCN cabling required of a redundant Safety Manager system.

In this case two SMM modules are installed, one each in the center and right-hand physical slots of a TRICON's logical slot. RG-6 UCN drop cables are run from the drop ports on UCN A to UCN A F-style connectors of each SMM. RG-6 UCN drop cables are also run from the drop ports on UCN B to UCN B F-style connectors of each SMM.

Figure 2-3 Redundant SM UCN Cabling



CAUTION Remove UCN cable from SMM port, without breaking UCN cable continuity, or communication will be lost to other UCN devices.

Continued on next page

2.6 Wiring and Cabling Installation, Continued

Power supply wiring

ATTENTION For information regarding the wiring of the Safety Manager card file power supply modules, refer to Subsection —*System Power Connections*, in this section.

I/O module wiring (Triconex TRICON Version 8)

Field wiring in the Triconex TRICON Version 8 is not connected directly to the I/O modules themselves. Rather, as illustrated in Figures 2-4 and 2-5, these connections are made via Field Termination Assemblies (FTA) available as either an

- Internal Termination Module which
 - is used for high-density applications,
 - resides in the left-most physical slot of the logical slot containing the I/O module(s), and
 - are specified for a given I/O module.
- External Termination Module which
 - permits remote terminal panel-based wiring connections,
 - connects via a cable bundle(s) to an interface module residing in the left-most physical slot of the logical slot, and
 - are specified for a given I/O module.

The FTAs are electrically passive devices which simply pass incoming or outgoing signals between the field and the I/O module. A separate Logical Slot Bus in each logical slot permits the I/O module(s) to communicate with the termination module.

ATTENTION Installation considerations specifically concerning power and ground connections can be located in Subsection 2.8—*System Ground Connections*.

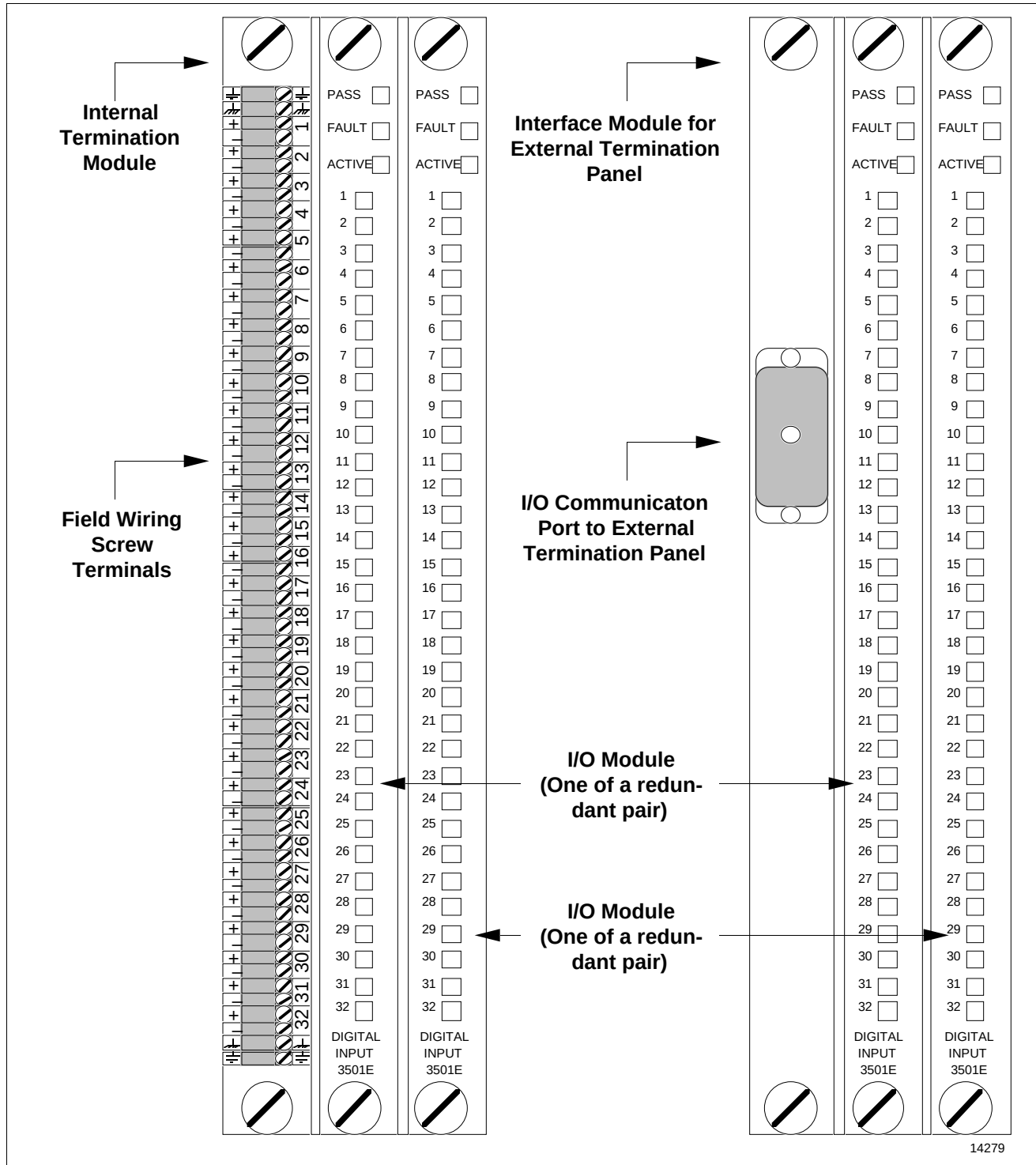
Figure 2-4 illustrates an example of the typical layout of two logical slots utilizing the Internal Termination Module and the Interface Module for the External Termination Module.

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2.6 Wiring and Cabling Installation, Continued

I/O module wiring,
continued

Figure 2-4 I/O Module Set Termination Examples (Triconex TRICON Version 8)



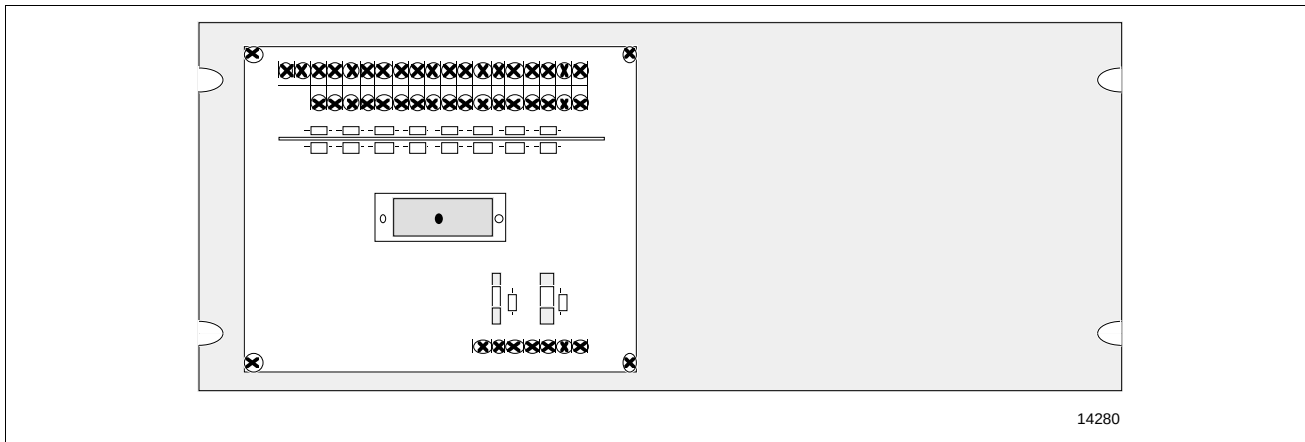
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2.6 Wiring and Cabling Installation, Continued

I/O module wiring, continued

Figure 2-5 illustrates an example of the type of External Termination Module employed by the TRICON.

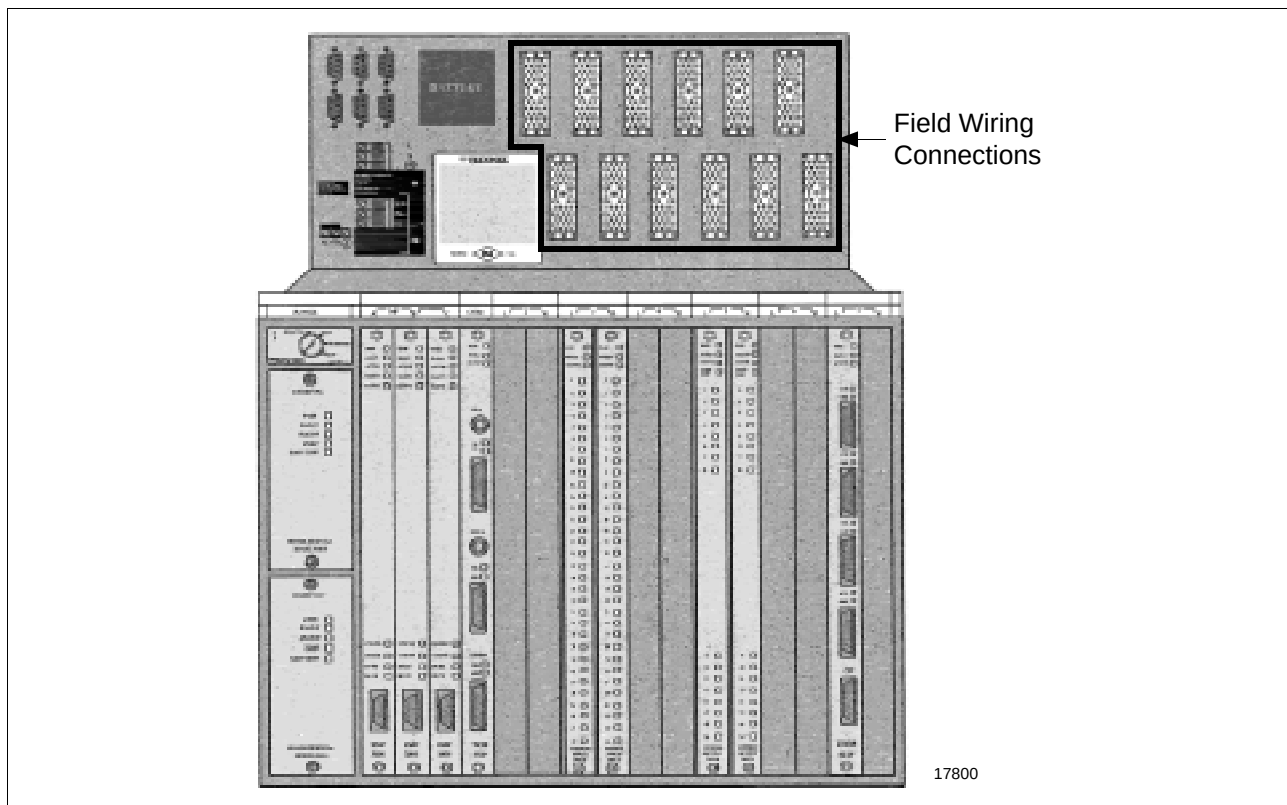
Figure 2-5 External Termination Module Example



I/O module wiring (Triconex TRICON Version 9)

Field wiring in the Triconex TRICON Version 9 is connected to the field wiring connectors that are mounted on the top of the backpanel as shown in Figure 2-6.

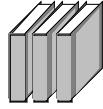
Figure 2-6 Main Chassis, Front View



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2.6 Wiring and Cabling Installation, Continued

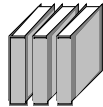
I/O module wiring (Triconex TRICON Version 9), Continued



Refer to the following references in your Triconex documentation.

- For I/O module specifications, refer to appropriate module in Chapter 3—*Selectable Modules*, in the *TRICON Planning and Installation Guide*.
- For field wiring precautions, refer to Chapter 4—*Installation and Checkout—Output Voter Diagnostics—Field Wiring Precautions*, in the *TRICON Planning and Installation Guide*.
- For recommended wiring methods, refer to Appendix F—*Recommended Wiring Methods*, in the *TRICON Planning and Installation Guide*.
- For field terminations, refer to the *User's Manual for Field Terminations*.

Communication module wiring

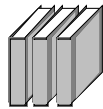


Refer to the following references in your Triconex documentation.

- For I/O module specifications, refer to appropriate module in Chapter 3—*Selectable Modules*, in the *TRICON Planning and Installation Guide*.
- For recommended wiring methods, refer to Appendix F—*Recommended Wiring Methods*, in the *TRICON Planning and Installation Guide*.

ATTENTION For information specific to the wiring of the Safety Manager Module, refer to “SMM Cabling” in this subsection.

TRISTATION cabling



Refer to the following references in your Triconex documentation.

- For TRISTATION cabling information, refer to
 - Chapter 4—*Installation and Checkout*, Subsection—*Connecting to the TRISTATION PC*, in the *TRICON Planning and Installation Guide*
 - Chapter 6—*Installation and Checkout*, Subsection—*Configuring the TRISTATION for RXM Chassis*, in the *TRICON Planning and Installation Guide*
 - Appendix D—*Pin-Outs for TRICON connectors*, Subsection—*TRICON-to-TRICON Cable*, and Subsection—*25-Pin to 9-Pin Adapter*, in the *TRICON Planning and Installation Guide*
- For recommended wiring methods, refer to the Appendix—*Recommended Wiring Methods*, in the *TRICON Planning and Installation Guide*.

2.7 System Power Connections

Cabinet power connections

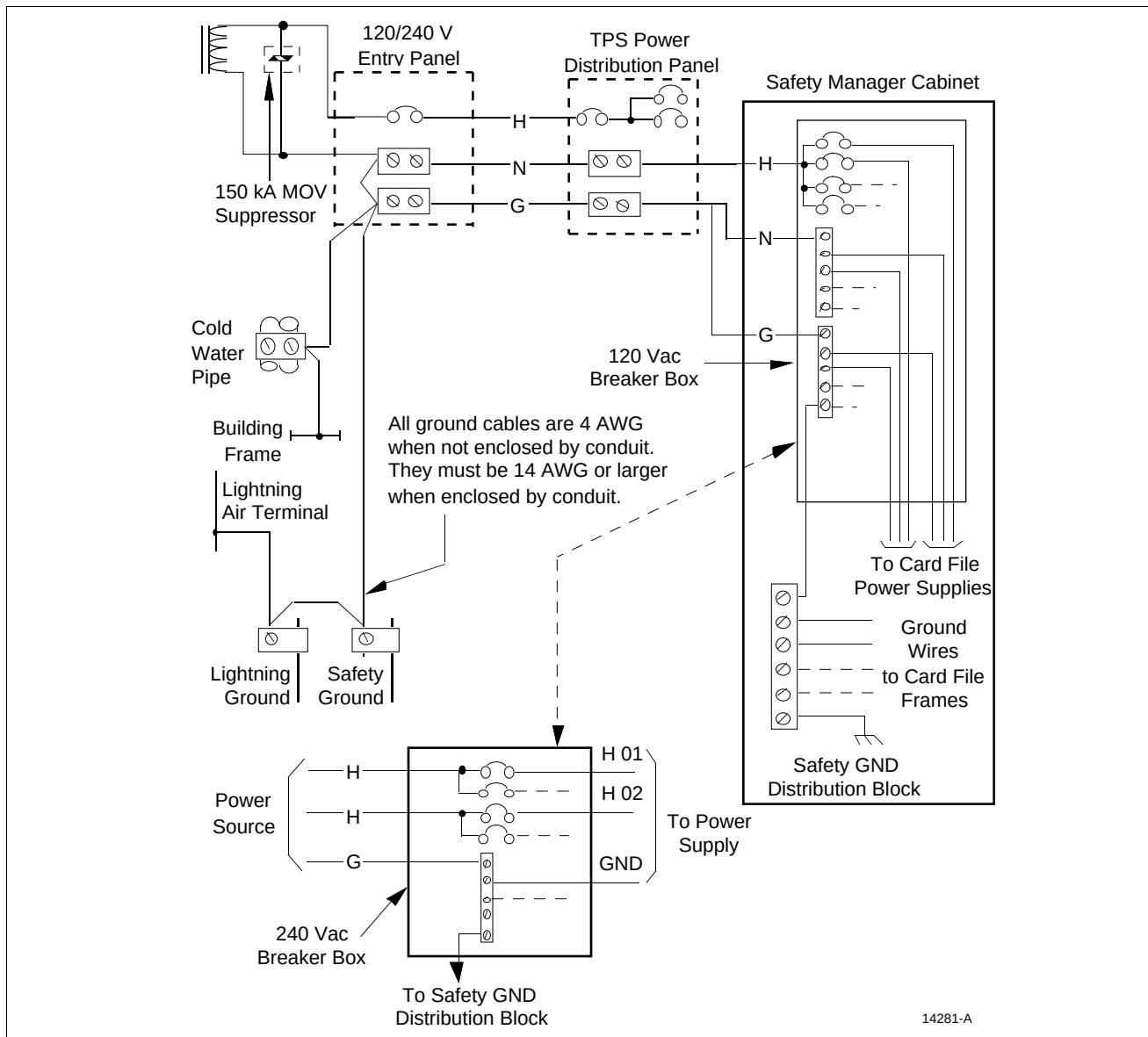
Figure 2-7 illustrates the power connections for a Safety Manager cabinet.

Conductors and breakers, used to supply power to the cabinet, are sized using rules in the *Site Planning* manual. Refer to your site planning drawings and this figure to install power.

A cabinet will have a circuit breaker box with either two or six dual 15-ampere circuit breaker boxes at the bottom of the cabinet. Each circuit breaker services a single card file.

CAUTION Do not use circuit breakers larger than 15 amperes; the power supply subassembly is not rated for larger circuit breakers.

Figure 2-7 Safety Manager Cabinet Power and Ground Connections

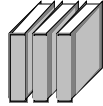


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2.7 System Power Connections, Continued

Card file power connections

Power is supplied to each Safety Manager card file (chassis) by connecting each power supply module in accordance to the specifications and techniques stipulated by Triconex.

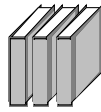


For information regarding the wiring of the Safety Manager card file power supply modules refer to the *TRICON Planning and Installation Guide*.

2.8 System Ground Connections

Background

The Safety Manager cabinets, card files (chassis), termination panels, communications equipment, and field wiring must be properly grounded to ensure correct and safe operation. This subsection provides the necessary information needed to make these connections.



For basic information regarding the TRICON's grounding requirements, refer to Chapter 4—*Installation and Checkout*, in the *TRICON Planning and Installation Guide*.

Grounding considerations

Adequate grounding is important for safety considerations and for reducing electromagnetic noise interference. All earth ground connections must be permanent and provide a continuous low impedance path to earth ground for induced noise currents and fault currents. Table 2-7 lists the steps to be performed when grounding your system.

Table 2-7 System Grounding Considerations

Step	Action
1	Use the system grounding techniques described in this subsection and those defined in the Triconex documentation before making ground connections. Establish • AC safety ground by making a – temporary connection using the grounding terminal on front of each power supply module, and – permanent, redundant connection to safety ground attaching No. 8 AWG from ground to each card file using a crimped ring lug and 1/4" bolt. • signal ground by tying together the signal ground and safety ground at <u>one</u> and <u>only one</u> point. • shield ground by connecting one end of the shield of each field cable between an analog termination panel and analog field device.
2	Select the proper voltage (115 Vac or 230 Vac) for all configurable modules before supplying power to the system.
3	Securely fasten all cover plates in order to maintain system integrity and to ensure system grounding.

Continued on next page

2.8 System Ground Connections, Continued

Grounding
considerations,
continued

Table 2-7 System Grounding Considerations, Continued

Step	Action
4	Ensure that cover plates are installed over all blank slots. (The cover plates supplied with Safety Manager system card files and modules shield the electronics from electromagnetic noise.)
5	Firmly tighten all thumbscrews on module front plates and cover plates.
6	Consult IEEE Standard 518-1977, the National Electric Code, and the Subsections 2.5 and 2.6 of this document before wiring the system components.

Cabinet ground
connections

Figure 2-7 illustrates the power and ground connections for a Safety Manager cabinet.

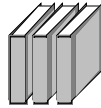
Ground wires from each card file in the cabinet are collected at a bus bar located at the bottom of the cabinet. This bus bar must be connected to building safety ground with a 1.67 mm (14 AWG) wire in conduit or a 5.3 mm (4 AWG) wire without conduit. Each cabinet must be connected individually to building safety ground and not daisy-chained with other cabinets or equipment.

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2.8 System Ground Connections, Continued

Safety ground

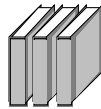
The Safety Manager card files (TRICON chassis) and external termination panels must be properly connected to safety ground in accordance to the specifications and techniques stipulated by Triconex.



For information on safety ground connections for the card files and the external termination panels, refer to the appropriate section in chapter 4 of the *TRICON Planning and Installation Guide*.

Signal ground

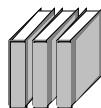
The Safety Manager card files (TRICON chassis) internal signal ground floats with respect to safety ground. To insure proper operation of the SM, signal ground must be configured in accordance to the specifications and techniques stipulated by Triconex.



For detailed information regarding the connection of signal ground for specific system configurations, refer to Chapter 4—*Installation and Checkout*.

Shield ground

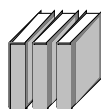
Field wiring between analog field devices for both the internal and external termination panels must be properly connected to shield (earth) ground. To ensure proper communication over these links, shield ground must be configured in accordance to the specifications and techniques stipulated by Triconex.



For detailed information regarding the connection of shield (earth) ground, refer to Chapter 4—*Installation and Checkout*.

Communication equipment grounding

Communication equipment connected to the Safety Manager must be grounded in accordance to the specifications and techniques stipulated by Triconex.



For information regarding the grounding of communication equipment, including the TRISTATION, refer to Chapter 4—*Installation and Checkout*.

2.9 UCN Connections

Node address selection	Unlike other UCN devices, the UCN node address is configured via the TRISTATION and becomes slot dependent.
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UCN installation	The installation procedure and cabling information can be found in the <i>UCN Site Planning and Installation</i> manual in the <i>UCN Site Planning and Installation</i> binder.
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Figure 2-8 illustrates a typical UCN installation. Connection of the UCN system is made at the front panel of the Safety Manager Module.

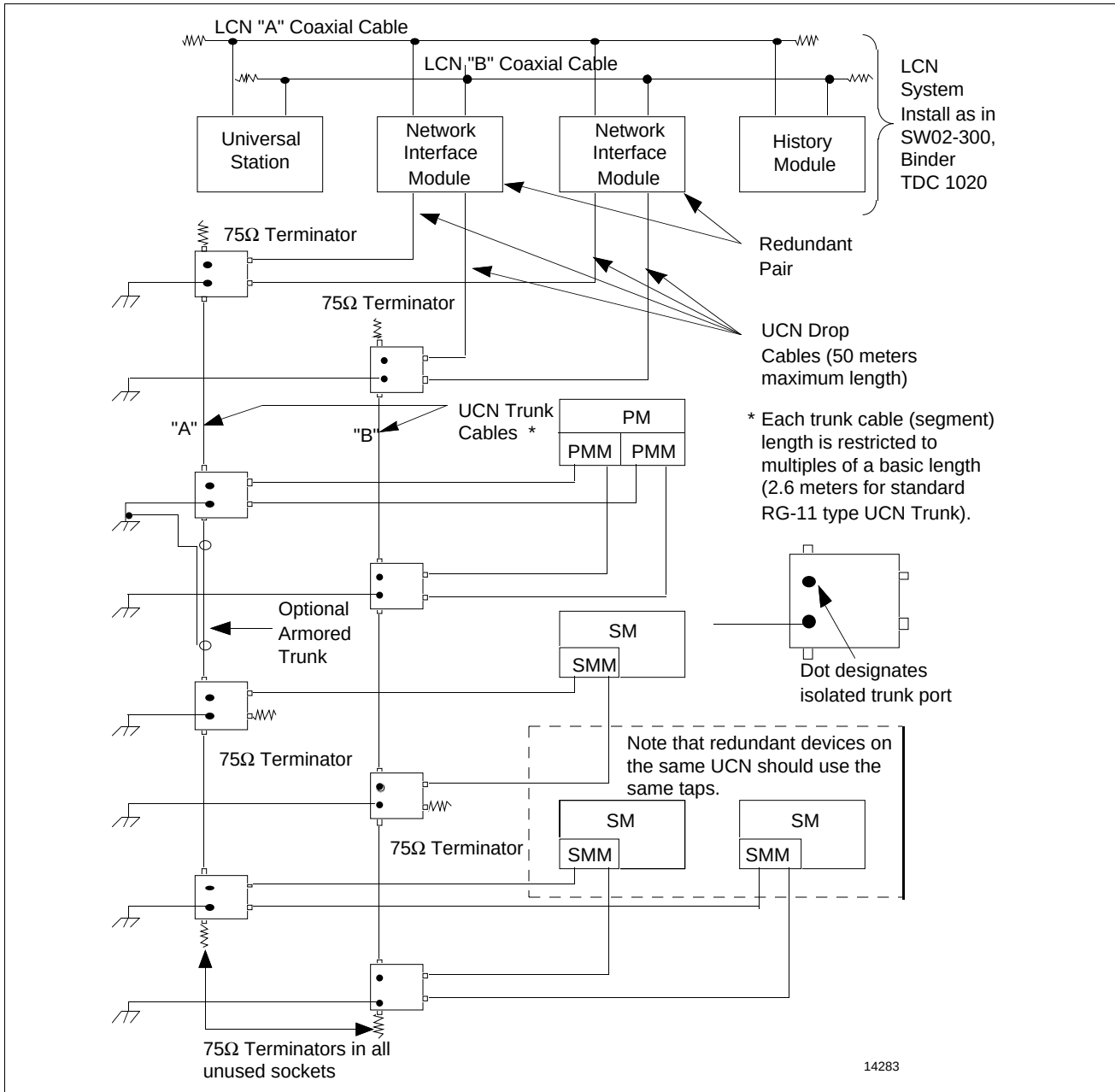
ATTENTION	For detailed information regarding the connection of the UCN to the Safety Manager, refer to Subsection 2.6— <i>Wiring and Cabling Installation</i> under the topic, “SMM Cabling,” in this document.
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2.9 UCN Connections, Continued

UCN installation, continued

Figure 2-8 Connection of the UCN



Trunk cable connection Connection of the UCN trunk lines are made at the front panel of the Safety Manager Module. Refer to Figure 2-3 for more detail. Use the procedure that is found in the *TPS System Site Planning* manual as the installation guide.

Section 3 – System Component Configuration

3.1 Section Overview

About this section

This section provides general installation information for various components of the Safety Manager (SM) system.

ATTENTION

This section describes in detail, only those installation procedures and considerations specific to the Safety Manager Module and the TPS installation procedures and considerations specific to the TRICON components can be located in the Triconex documentation.

3.2 System Configuration Overview

Introduction

The Safety Manager (SM) system consists of a TRICON Programmable Logic Controller (PLC) outfitted with a single or redundant pair of Safety Manager Modules (SMMs).

The TRICON's main chassis card file contains the PLC's triplicated main processors and some mix of I/O and communications modules. Typically, the SMM(s) also reside in the main chassis, but may be located in either an expansion or RXM chassis. The SM also supports a dual-redundant SMM configuration for load dividing.

Redundant SMM modules

The SM can be configured for either non-redundant or redundant operation depending on the number of SMM modules employed. In a redundant configuration, two SMMs are installed in adjacent physical slots of any compatible logical slot.

In this configuration, each SMM is independently connected to the UCN. One of the two will function as the primary, carrying out all UCN to SM communication and data transfer operations. The other SMM will function as the secondary or backup link between the UCN and TRICON. A separate private link is established between the two SMMs via the logical slot's independent backplane. Database synchronization and status and control transfers occur over the private link; the UCN is used as backup.

3.3 Card File Configuration

Introduction

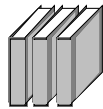
The Safety Manager has three chassis categories which are as follows:

- Main chassis
 - Expansion chassis
 - Remote expansion chassis
-

Main chassis

The main chassis (card file) includes several standard items such as the redundant chassis power supplies, keyswitch, and Main Processor modules. It also has four logical slots (excluding those reserved for the main processors) in which can be installed:

- communication modules (such as the SMM),
- I/O modules,
- hot-spare I/O modules, and
- termination modules.



Refer to the following references in your Triconex documentation for detailed main chassis configuration requirements.

- For modules/slot compatibility and configuration, refer to Chapter 2—*Basic Components*, Section—*Main and Expansion Chassis*, in the *TRICON Planning and Installation Guide*.
 - For information regarding the configuration of the I/O module identification keys, refer to Chapter 4—*Installation and Checkout*, Section—*Recommended Installation Procedures*, Subsection—*I/O Module Identification Keys* in the *TRICON Planning and Installation Guide*.
 - For information regarding the configuration of the chassis I/O bus address, refer to
 - Chapter 4—*Installation and Checkout*, Section—*Recommended Installation Procedures*, Subsection—*I/O Bus Address of TRICON Chassis* in the *TRICON Planning and Installation Guide*
 - Configuration using the TRISTATION. Refer to the *TRISTATION MSW User's Manual*
-

Expansion chassis

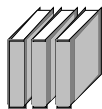
The expansion chassis includes I/O modules and its own set of redundant power supplies. Each chassis has six RS-485 I/O expansion bus ports for passing data to two other chassis.

Continued on next page

3.3 Card File Configuration, Continued

RXM chassis

The fiber-optic remote extender module (RXM) is for operation of expansion chassis up to 2 kilometers (1.2 miles) away from the main chassis.



For more information on expansion and RXM chassis, refer to Triconex's *TRICON Planning and Installation Guide*.

Section 4 – Configuration

4.1 Section Overview

About this section

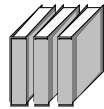
This section provides general information for various software configuration procedures for the Safety Manager (SM) system.

Introduction

There are four areas which need to be configured to properly use the Safety Manager. These are as follows:

- TRICON relay ladder logic programming,
- TRISTATION configuration for the UCN,
- Sequence of Events (SOE) configuration, and
- Universal Station configuration for SMM points.

This section will focus primarily on the TRISTATION configuration for UCN, SOE configuration and US configuring of SMM points.

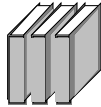


For information on TRICON relay ladder logic programming, refer to the Triconex *TRISTATION MSW User's Manual*.

Preconfiguration assumptions

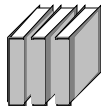
This section assumes the following steps have been completed:

- The TRISTATION hardware (i.e. cabling) has been properly connected.



Refer to wiring section of this manual.

- The TRISTATION software has been successfully loaded onto the PC.



Refer to the Triconex *TRISTATION MSW User's Manual*.

Version needed for configuration

In order to successfully configure the SM system, you will need to be using version 9 of Triconex TRICON software and release 610 for the TPS system.

4.2 TRISTATION UCN Configuration

Configuring for the UCN at the TRISTATION

In order to ensure that the Safety Manager is synchronized with other devices on the UCN and its SOE capabilities are available, the SM must have the UCN address configured at the TRISTATION. Tables 4-1 and 4-2 outline this procedure. Use Figure 4-1 as a visual reference.

Table 4-1 Getting to the TRISTATION UCN Configuration Screen

Step	Action
1	Bring up your TRICON software by typing <code>ts32</code> on your PC. The FILE MANAGER display will be the first screen you view after entering the correct password level. ATTENTION Make sure the [NUM LOCK] key is off first.
2	Press the [ALT 3] keys. You should now be in the MODULE CONFIGURATOR screen.
3	Determine your desired SMM configuration in chassis. ATTENTION It is possible to have two redundant (four) SMMs configured for a Safety Manager system. SMMs can be configured for slots 5, 6 and 9, 10 in chassis 1 or chassis 2. There are configuration restrictions which apply.
4	Use the [F7 PREV SLOT] and [F8 NEXT SLOT] keys to move from previous to next slots in the chassis. Once you have reached the slot you wish to configure, press the [F6 CONFIG MODULE].
5	Use the arrow keys to locate and highlight SMM under Module Selections on the screen. Click on the [F1 ADD MODULE] key. An SMM configuration window will come up.
6	Using the arrow keys to highlight the parameter you want, configure as desired and press the [ENTER] key. ATTENTION Table 4-2 lists the configuration parameters in detail.
7	Press the [ESC] key to exit out of configuration window once configuration is complete.

Continued on next page

4.2 TRISTATION UCN Configuration, Continued

Module Configurator screen

Figure 4-1 shows you the Module Configurator screen for the TRISTATION.

Figure 4-1 TRISTATION Module Configurator Screen

MODULE CONFIGURATION		Discrete Input, 115V, 32 point Model 3501/3501E																																							
F1 ADD MODULE	F2 DELETE MODULE	■ = TSS600 MODULE SELECTIONS ◆ = simplex																																							
		EITC T dgF DnS			EITC E dgF DnS			EITC J dgC DnS			EITC K dgF DnS																														
		EITC T dgC UpS			EITC E dgC			EITC J dgF UpS			EITC K dgC UpS																														
F3 ADD CHAS	F4 CHAN/ DEL CH	EITC T dgF UpS			UpS			AIO 10V			AIO 5V																														
		AO 4-20 MA			EITC E dgF UpS			HIM			PULSE INPUT																														
		DCM/NCM			EICM			SMM																																	
F5 PRINT	F6 CONFIG MODULE	■ = TSS600 CHASSIS CONFIGURATION ◆ = simplex																																							
		1 2 3 4																																							
		<table><tr><td>MP</td><td>MP</td><td>MP</td><td>SMM</td><td></td><td></td><td></td><td></td><td>SMM</td><td>ICM</td><td></td><td></td><td></td><td></td></tr><tr><td>1</td><td>2</td><td>3</td><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td><td>10</td><td>11</td><td>12</td><td></td><td></td><td></td></tr></table>												MP	MP	MP	SMM					SMM	ICM					1	2	3	5	6	7	8	9	10	11	12			
MP	MP	MP	SMM					SMM	ICM																																
1	2	3	5	6	7	8	9	10	11	12																															
F7 PREV SLOT	F8 NEXT SLOT																																								
F9 PREV CHAS	F2 NEXT CHAS	Chassis 1 of 2 Type: Lo Density Main - TMR																																							

Help: <ALT> = Menu <ALT> M Program: SMMCONFG

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Continued on next page

4.2 TRISTATION UCN Configuration, Continued

TRISTATION UCN configuration parameters

Table 4-2 lists the TRISTATION UCN configuration parameters and their corresponding functional descriptions.

Table 4-2 TRISTATION UCN Configuration Parameters

Parameter	Description
Node address	This is the address of the SM node on the UCN. This number <u>must</u> match the node number assigned when using the US to configure SMM data points for the same SM device. Enter in an odd number from 1 to 63. ATTENTION Odd numbered addresses are reserved for active modules and even addresses for hot spare modules. Up to 32 redundant devices are supported by the UCN. Dual redundant SMMs should have different UCN addresses.
SOE block	You must answer with 15 or 16 if you wish to collect SOE data from the TRICON. ATTENTION SOE blocks 15 and 16 are for the SMM only. Note that if you have two redundant SMMs configured in the same chassis, they <u>cannot</u> be the same SOE block number. If one is 15 the other must be 16.
Prohibit writes	Answer NO if you wish to allow writes to be made to the TRICON from the Universal Station. If not, answer YES.
UCN test mode	Answer NO to disable factory test mode. ATTENTION A YES answer should be used only by Honeywell factory test personnel.
MP Clock	An answer YES means you want this pair of SMMs to update the TRICON with the UCN time. You must answer YES if you plan to configure SMM digital input points for use with the SOE application. ATTENTION If you have two redundant SMMs configured on the same chassis, you can answer YES for only <u>one</u> of the logical pairs.

Continued on next page

4.2 TRISTATION UCN Configuration, Continued

Saving TRISTATION UCN configuration data

Once you have completed your TRISTATION UCN configuration scheme, you must save this program in your database. To do this, follow steps outlined in Table 4-3. Use Figure 4-2 as a visual reference.

Table 4-3 Saving and Downloading TRISTATION UCN Data

Step	Action
1	Once your configuration is complete, call up the FILE MANAGER screen. To do this press the [ALT 0] keys.
2	Press the [F2 SAVE PROGRAM] key to save the configuration you have just completed. It will prompt you for a file name.
3	Enter in a file name for your configuration program and press the [ENTER] key. You have now saved your program and should see it on your screen.
4	Download the configuration data with the [F10 DOWNLD ALL] function key.

File Manager screen

Figure 4-2 shows the File Manager screen for the TRISTATION.

Figure 4-2 TRISTATION File Manager Screen

FILE MANAGEMENT		
F1 LOAD PROGRM	F2 SAVE PROGRM	TRISTATION MSW 2.0.1 (4507, 4518) Free disk space: 20.983 Mb Current version of "SOECONFG":1.1 R11 files available in "C:\MSW200": SMMCONF SOECONFG
F3 NEW PROGRM	F4 EXIT	
F5 DELETE FILE	F6 CHANGE DIR	
F7 LIST FILES	F8 LIST OTHER	
F9 DOWNLD CHANGE	F2 DOWNLD CONFIG	
		Help: <ALT> = Menu <ALT> M Program: SOECONFG

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Continued on next page

4.2 TRISTATION UCN Configuration, Continued

How to tell if the SM is configured

Once configuration is complete, the SMM's 4-character alphanumeric display will show A and the node number configured for the active module, and A and the associated even address number for the backup (hot spare) module.

If the SMM(s) is not yet configured, the alphanumeric display windows on the Safety Manager Modules will both show CFG? The PI cannot be loaded.

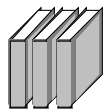
ATTENTION

While performing a DOWNLOAD ALL, the TRISTATION displays may report the SMM(s) as missing. This is normal if the download involved a configuration change that forced the SMM(s) to restart, thus making them temporarily blind to the Main Processors. Once an SMM completes restart, it will automatically re-establish a connection with the TRICON Main Processors.

4.3 Sequence of Events (SOE) Configuration

Configuring your system for SOE

During each scan of a control program, the TRICON examines selected discrete variables for state changes known as events. SOE block configuration allows you to specify which discrete aliased variables defined in the control program are to be monitored by the TRICON for state changes.



Refer to Triconex's *Sequence of Events User's Manual* for detailed information about SOE functions and capabilities.

Setting up the control program for SOE

To use SOE, you must configure the TRISTATION to do the following when setting up your control program:

- Define the event variables and their alias numbers in the Dictionary Editor screen [ALT 2].

ATTENTION Event variables must be of the DI type.

- Add SOE functions to your ladder logic to start and stop the collection of data into your SOE blocks using the Ladder Editor screen [ALT 1].

ATTENTION Special functions are provided for this and can enable either SOE blocks 15, 16 or both. These functions should only be called on the first TRICON scan, using the TRICON “first scan” condition in the ladder logic program.

- Download the control program with the [F10 DOWNLD ALL] function of the File Manager screen [ALT 0].

This subsection will focus on grouping the event variables using the SOE Block Configuration screen. Refer to the Triconex *SOE User's Manual* for other control program manipulation for enabling SOE.

About SOE block configuration

There are a total of 16 SOE blocks. Blocks 15 and 16 are always External Blocks for use with the Safety Manager Module(s) only. All other discrete variables are defined at the Universal Station. These are the only blocks you will be configuring.

SOE Block Configuration screen

The SOE Block Configuration screen allows you to organize event variables for your control program as blocks. When you configure an External Block, block size and name are the only items specified. DI points to be collected are configured at the US.

Alias range limitations

Only alias ranges 2001-4000, 10000-12000, and 12001-14000 are available for DISOE. The PLCADDR parameter of a DI point must be within the above ranges before SOE can be enabled for a given point.

Continued on next page

4.3 Sequence of Events (SOE) Configuration, Continued

Using the SOE Block Configuration screen

Use Table 4-4 to guide you through the SOE block configuration procedure and Figure 4-3 as a visual reference.

Table 4-4 SOE Block Configuration Procedure

Step	Action
1	Press the [ALT 4] keys to bring up the SOE Block Configuration screen.
2	Use the [F1 PREV BLOCK] and [F2 NEXT BLOCK] keys to get to block 15 or 16.
3	Press the [F3 CONFIG BLOCK] key. A window for SOE configuration parameters will come up.
4	Use the arrows and [ENTER] key to configure your SOE parameters as desired. ATTENTION Blocks 15 and 16 are used exclusively for the SMM. Buffer size and name will be the only items configured for these blocks. DI points to be collected are configured at the US.
5	Press the [ESC] key once you have completed your configuration.

SOE Block Configuration screen

Figure 4-3 shows the SOE Block Configuration screen.

Figure 4-3 TRISTATION SOE Block Configuration Screen

SOE BLOCK CONFIGURATION		Title: SM SOE CONFIG	Buffer: 100 events
F1 PREV BLOCK	F2 NEXT BLOCK	BLOCK 15/16 - MOD_EXTERNL 0 SYMBOLS HAVE BEEN SELECTED	
F3 CONFIG BLOCK	F4 CLEAR BLOCK	SELECT-NAME DESCRIPTION ALIAS	
F5 SELECT ON/OFF	F6 CLEAR SELECT	CONFIGURE SOE BLOCK PARAMETERS Title : SM SOE CONFIG Buffer Size: 100 (maximum is 20000 Type : events) MOD_EXTERNL	
F7 FIND NAME	F8 FIND ALIAS		
F9 VIEW OPTION	F2 PRINT CONFIG		

Help: <ALT> = Menu <ALT> M Program: SOECONFG

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Continued on next page

4.3 Sequence of Events (SOE) Configuration, Continued

Saving and downloading SOE data

Once you have completed your SOE configuration scheme, you must save and load this program into your database. To do this, follow the steps outlined in Table 4-5. Refer back to Figure 4-2 as a visual reference.

Table 4-5 Saving and Downloading SOE Configuration Data

Step	Action
1	Once your configuration is complete, call up the FILE MANAGER screen. To do this press the [ALT 0] keys.
2	Press the [F2 SAVE PROGRAM] key to save the configuration you have just completed. It will prompt you for a file name.
3	Enter in a file name for your SOE configuration program and press the [ENTER] key. You have now saved your program and should see it on your screen.
4	Download the configuration data with the [F10 DOWNLD ALL] function key.

4.4 US Configuration

Introduction

Once the UCN has been configured at the TRISTATION, you will need to use the Universal Station (US) to complete SM system configuration. This section will discuss the following US configurations:

- Universal Control Network (UCN),
- Node-specific (SM device), and
- SMM data points.

Configuration assumptions

This section covers the SM configuration process at the US. It assumes that you have completed the implementation steps for start-up and have prepared the History Module (HM) and the system for network configuration.

Table 4-6 outlines the start-up process for the TPS system. If these procedures have not been done, refer to the TPS bookset - *Implementation Startup & Reconfiguration*, binder TPS 2030-1.

Table 4-6 Start-up and Reconfiguration Procedures

Step	Action	Result
1	Prepare the system for network configuration.	Engineer Personality is in "this US." Current time and date is in "this US." The system start-up floppy is in a secure place.
2	Configure the network.	A site-specific Network Configuration File (NCF), required as input to subsequent procedures, is available.
3	Prepare the History Module.	The History Module can now be used as a source for configuration programs and data.

UCN configuration

Once you have completed the start-up process for the US, follow the steps outlined in Table 4-7.

Table 4-7 UCN Configuration

Step	Action
1	Press the RESET button. ATTENTION Reset button is located under black metal panel at top of operator keyboard.
2	Press the LOAD button on operator keyboard. System will state 'Loading in progress'.
3	A prompt will come up "N, 1, 2, 3, 4?", press N for Node and press the [ENTER] key.
4	Another prompt will come up "OPR, UNP, X?", press U for Universal Personality and press the [ENTER] key.

Continued on next page

4.4 US Configuration, Continued

First US display

Once you have loaded the Universal Personality, the System Status display will be the first display to come up. Figure 4-4 shows this display.

Figure 4-4 US System Status Display

MAKE SELECTION

02 May 95 11:13:09 3

SYS VERS: R610

SYSTEM STATUS FROM US13

NCF VERS: 09Feb95 00:14:19:157

CABLE A: OK CABLE B: OK

US01	US02	US03	US04	US05	US06	US07	US08	US09	US10	US11	US12	US13
OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OK
US14	US15	US16	US17	US18	US19	US20	NM21	NM22	NM23	NM24	NM25	NM26
OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OK	OFF	OFF
NM27	NM28	NM29	NM30	NM31	HG37	HG39	HM43	AM45	AM46	CG 49		
OFF	OFF	OFF	OFF	OFF	OFF	OFF	OK	OFF	OFF	OFF		

LOAD SELECT

LOAD ISOL

LOAD CONFIG

LOAD FAIL

SHOW LOADS

LOAD PWR ON

DE-SELECT

SELECT AUX INFO

NTWK/HWY STATUS

DUMP NODE

NODE STATUS

SHUT DOWN

STATUS DETAIL

LCN OVERVIEW

LCN DIAG

TIME/DATE

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Engineering Main Menu display

In order to configure your UCN, you will need to bring up the Engineering Main Menu display. To do this, hold the [CTRL] key on the engineering keyboard and press the [HELP(M)] key. Figure 4-5 shows this display.

Once you have brought this display up, select the NETWORK INTERFACE MODULE target.

Figure 4-5 US Engineering Main Menu Display

02 May 95 13:19:57 3

ENGINEERING MAIN MENU

UNIT NAMES

AREA NAMES

CONSOLE NAMES

LCN NODES

SYSTEM WIDE VALUES

VOLUME CONFIGURATION

HIWAY GATEWAY

LOGIC BLOCKS

APPLICATION MODULE

COMPUTING MODULE

NETWORK INTERFACE MODULE

BUILDER COMMANDS

PICTURE EDITOR

FREE FORMAT LOGS

BUTTON CONFIGURATION

HM HISTORY GROUPS

DOCUMENTATION TOOL

AREA DATA BASE

Support Functions and Utility Programs

COMMAND PROCESSOR

SYSTEM MENU

SYSTEM STATUS

CONSOLE STATUS

SUPPORT UTILITIES

SMCC/ MAINTENANCE

RXXX (C) HONEYWELL INC., 1984

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Continued on next page

4.4 US Configuration, Continued

NIM Build Type Select Menu display

The NIM Build Type Select Menu display will be the next display to come up. This display is shown in Figure 4-6. Select the UCN NODE CONFIGURATION target to configure the UCN.

Figure 4-6 US NIM Build Type Select Menu

02 May 95 13:22:07 3

MENU

PAGE 01 OF 01

NIM BUILD TYPE SELECT MENU

UCN
NODE
CONFIGURATION

NODE
SPECIFIC
CONFIGURATION

PROCESS
POINT
BUILDING

LIBRARY
CONFIGURATION

F1=PED

F3=

F5=OVERWRITE

F7=RECON

F9 =WLK BACK

F11=

F2=RECALL DISP

F4=

F6=

F8=PED STATUS

F10=WRITE

F12=LOAD

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UCN Node Configuration display

You should now be in the UCN Node Configuration display. Figure 4-7 shows this display. Do the steps in Table 4-8 to configure the UCN.

Table 4-8 UCN Node Configuration

Step	Action
1	Enter the requested data in the given slots. All configuration information should be taken from the SM Configuration Forms you have filled out.
2	Select the SM for UCN node type and TRICON for the SM platform.
3	Press the [F12 LOAD] key to load your configuration.

ATTENTION The node number you select must correspond with the node number you entered into the TRISTATION UCN configuration.

Continued on next page

4.4 US Configuration, Continued

UCN Node Configuration display, continued

Note that the network number can be from 1 to 20. The node number can range from 1 to 63, using odd numbers only.

Figure 4-7 US UCN Node Configuration Display

02 May 95 13:25:17 3

PED >>>>>> POINT:\$NM02N13

UNIT:SY

PAGE 01 OF 01

UCN NODE CONFIGURATION

NETWORK NUMBER

(NTWKNUM)

02

NODE NUMBER

(NODENUM)

13

UCN NODE TYPE

(NODETYP)

NOTCONFG

NIM

PM

APM

HPM

LM

SM

PM1

SM PLATFORM

(SMPLTFM)

TRICON

FSC

NODE ASSIGNMENT

(NODEASSN)

THISNIM

REMOTNIM

DIGITAL COMPOSITE PV
STATE TEXT

MOVING TEXT

(MOVPVTXT)

MOVING

BAD TEXT

(BADPVTXT)

BAD

F1=PED

F3=

F5=OVERWRITE

F7=RECON

F9 =WLK BACK

F11=

F2=RECALL DISP

F4=

F6=

F8=PED STATUS

F10=WRITE

F12=LOAD

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Continued on next page

4.4 US Configuration, Continued

Node-specific configuration

To configure the node-specific data, return to the NIM Build Type Select Menu by pressing the [F9 WLK BACK] key. Refer back to Figure 4-6. Select the NODE SPECIFIC CONFIGURATION target. The Node-Specific Configuration display is shown in Figure 4-8.

Enter and select the desired configuration. Press the [F12 LOAD] key.

ATTENTION Make sure the network and node numbers entered for this display match those entered in the UCN Configuration display.

There is no need to configure the redundant SM device. All SM-related information will be automatically transferred to the “new” primary SM should there be a failover or switchover to the backup.

Figure 4-8 US Node-Specific Configuration Display

02 May 95 13:29:30 3

PED >>>>>> POINT:\$NM02B13UNIT:SYUNENTEREDPAGE 01 OF 03

NODE-SPECIFIC CONFIGURATION

NETWORK NUMBER (NTWKNUM) 02

NODE NUMBER (NODENUM) 13

NODE TYPE (NODETYP) PMAPMHPMLMSM^{PM1}

SM PLATFORM (SMPLTFM) TRICONFSC

*
* WARNING: IF THE POINT MIX OR THE SCAN
* RATE IS CHANGED, THEN THE NODE'S FULL
* DATA BASE WILL BE SET TO ITS DEFAULT.
* THE DISPLAY LINES WHICH CAN CAUSE THE
* DATA BASE TO BE DEFAULTED ARE STARRED
* AND HAVE AN ASTERISK (*) IN COLUMN 80.
*

F1=PEDF3=F5=OVERWRITEF7=RECONF9 =WLK BACKF11=
F2=RECALL DISP F4=F6=F8=PED STATUSF10=WRITEF12=LOAD

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Continued on next page

4.4 US Configuration, Continued

Process Point Building display

You must once again return to the NIM Build Type Select Menu to access the SMM data point building displays. To do this, press the [F1 PED] key.

Select the PROCESS POINT BUILDING target, once you have returned to the NIM Build Type Select Menu. Figure 4-9 shows the display which will be brought up.

Figure 4-9 US Process Point Building Display

02 May 95 13:33:40 3

MENU

PAGE 01 OF 01

NIM BUILD TYPE SELECT MENU

NIM PROCESS POINT BUILD TYPE

ANALOG INPUT

DIGITAL INPUT

REGULATORY PV

LOGIC

FLAG

ARRAY

PROCESS MODULE

ANALOG OUTPUT

DIGITAL OUTPUT

REGULATORY CONTROL

TIMER

DEVICE CONTROL

DIGITAL COMPOSITE

NUMERIC

F1=PED

F2=RECALL DISP

F3=

F4=

F5=OVERWRITE

F6=

F7=RECON

F8=PED STATUS

F9 =WLK BACK

F10=WRITE

F11=

F12=LOAD

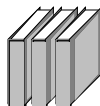
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Continued on next page

4.4 US Configuration, Continued

Building SMM data points

This subsection takes you through the general steps for SMM data point configuration. To build SMM data points, you will need to follow the steps listed in Table 4-9.



Refer to Appendix A for a page-by-page view of the configuration parameters for a given SMM point type. Also refer to *Section 4—Data Point Detailed Description* in document SM09-600, *Safety Manager Module Control Functions*, for more information.

ATTENTION Information on the definition of a specific parameter can be provided to you online by pressing the [HELP] key for the parameter you have selected.

Table 4-9 SMM Data Point Configuration Procedure

Step	Action
1	Once you are in the Process Point Building display, select the SMM point type you wish to configure.
2	Enter and select the desired configuration scheme. Use the [PAGE FWD] button on the operator keyboard to move to the next configuration page as needed. ATTENTION Data entered should be based on the SM Configuration Forms you have filled out.
3	Press the [F12 LOAD] key to load the data in the SM database.
4	Press the [F9 WLK BACK] key to return to the Process Point Building display.
5	Repeat steps for next SMM data point type until all point types are configured.

ATTENTION TRICON aliases must be configured before points can be downloaded against given aliases.

Appendix A – SMM Data Point Configuration Displays

A.1 Overview

About this section

This appendix contains sample displays that can be viewed from the Universal Station when configuring SMM data points.

ATTENTION The highlighted selections in these displays are for example purposes only. Make your selections based on your process needs.

A.2 Analog Input

Page 1 of 7

26 Jul 99 15:28:22 4

PED >>>>>> POINT:		UNIT:	ERRS UNENTERED	PAGE 01 OF 07
NIM-POINT ASSIGNMENT ANALOG INPUT				
TAG NAME	(NAME)			
NODE TYPE	(NODETYP)	PM	APM	HPM
		LM	SM	
SM PLATFORM	(SMPLTFM)	TRICON	FSC	
POINT FORM	(PNTFORM)	FULL	COMPONNT	
POINT DESCRIPTOR	(PTDESC)			
E.U. DESCRIPTOR	(EUDESC)			
POINT KEYWORD	(KEYWORD)			
ASSOCIATED DISPLAY	(ASSOCDSP)			
UNIT ID	(UNIT)	☐		
NETWORK NUMBER	(NTWKNUM)	☐		
F1=PED F3= F5=OVERWRITE F7=RECON F9 =WLK BACK F11=TAB F2=RECALL DISP F4= F6= F8=PED STATUS F10=WRITE F12=LOAD				

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Page 2 of 7

26 Jul 99 15:29:43 4

PED >>>>>> POINT:		UNIT:	ERRS	PAGE 02 OF 07
NODE NUMBER	(NODENUM)	☐		
SLOT NUMBER	(SLOTNUM)			
PRIMARY MODULE POINT ID	(PRIMMOD)	-----		
F1=PED F3= F5=OVERWRITE F7=RECON F9 =WLK BACK F11=TAB F2=RECALL DISP F4= F6= F8=PED STATUS F10=WRITE F12=LOAD				

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Continued on next page

A.2 Analog Input, Continued

Page 3 of 7

26 Jul 99 15:30:514

PED >>>>>> POINT:

UNIT:ERRS

PAGE 03 OF 07

SM SOURCE ADDRESS

(PLCADDR)

0

F1=PED

F2=RECALL DISP

F3=

F4=

F5=OVERWRITE

F6=

F7=RECON

F8=PED STATUS

F9 =WLK BACK

F10=WRITE

F11=TAB

F12=LOAD

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14210-A

Page 4 of 7

26 Jul 99 15:33:17 4

PED >>>>> POINT:		UNIT:	ERRS	PAGE 04 OF 07	
NIM-PV CONFIGURATION					
ANALOG INPUT					
CHARACTERIZATION	(PVCHAR)	LINEAR SQRRROOT TC_RTD			
PV RAW HIGH LIMIT	(PVRAWHI)	1.0			
PV RAW LOW LIMIT	(PVRAWLO)	0.0			
INPUT DIRECTION	(INPTDIR)	DIRECT REVERSE			
PV RANGE HIGH	(PVEUHI)	100.0			
PV RANGE LOW	(PVEULO)	0.0			
PV DECIMAL FORMAT	(PVFORMAT)	00	01	02	03
PV RANGE EXTENSION HIGH	(PVEXEUHI)	102.9			
F1=PED F3= F5=OVERWRITE F7=RECON F9 =WLK BACK F11=TAB F2=RECALL DISP F4= F6= F8=PED STATUS F10=WRITE F12=LOAD					

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Continued on next page

A.2 Analog Input, Continued

Page 5 of 7

26 Jul 99 15:34:28 4

PED >>>>>> POINT:		UNIT:	ERRS	PAGE 05 OF 07
PV RANGE EXTENSION LOW	(PVEXEULO)	-2.9		
PV CLAMPING OPTION	(PVCLAMP)	CLAMP	NOCLAMP	
LOW SIGNAL CUTOFF	(LOCUTOFF)	-----		
PV SOURCE OPTION	(PVSRCOPT)	ONLYAUTO	ALL	
PV FILTER LAG TIME	(TF)	0.0		
PV TARGET VALUE	(PVTV)	-----		
OVERVIEW VALUE	(OVERVAL)	25		
F1=PED F3= F5=OVERWRITE F7=RECON F9 =WLK BACK F11=TAB F2=RECALL DISP F4= F6= F8=PED STATUS F10=WRITE F12=LOAD				

17801

Page 6 of 7

26 Jul 99 15:35:16 4

PED >>>>>> POINT:		UNIT:	ERRS	PAGE 06 OF 07
NIM-ALARMING DISPLAY ANALOG INPUT				
AUXILIARY UNIT	(\$AUXUNIT)	--		
PV ALARM DEADBAND IN %	(PVALDB)	HALF	ONE	TWO THREE FOUR FIVE EU
PV HIGH TRIP POINT	(PVHITP)	----		
PV LOW TRIP POINT	(PVLOTP)	----		
BAD PV ALARM PRIORITY	(BADPVPR)	NOACTION	JOURNAL	LOW HIGH EMERGNCY PRINTER JNLPRINT
ALARM ENABLE STATE	(ALENBST)	ENABLE	DISABLE	INHIBIT
F1=PED F3= F5=OVERWRITE F7=RECON F9 =WLK BACK F11=TAB F2=RECALL DISP F4= F6= F8=PED STATUS F10=WRITE F12=LOAD				

17802

Continued on next page

A.2 Analog Input, Continued

Page 7 of 7

26 Jul 99 15:36:094

PED >>>>> POINT:UNIT:ERRSPAGE 07 OF 07

CONTACT CUTOUT SOURCE(CCSRC)0

F1=PEDF3=F5=OVERWRITEF7=RECONF9 =WLK BACKF11=TAB

F2=RECALL DISPF4=F6=F8=PED STATUSF10=WRITEF12=LOAD

17803

A.3 Analog Output

Page 1 of 5

26 Jul 99 15:46:18 4

PED >>>>>> POINT:		UNIT:	ERRS UNENTERED	PAGE 01 OF 05
NIM-POINT ASSIGNMENT ANALOG OUTPUT				
TAG NAME	(NAME)			
NODE TYPE	(NODETYP)	PM	APM	HPM
		LM	<input checked="" type="text"/> SM	
SM PLATFORM	(SMPLTFM)	<input checked="" type="text"/> TRICON	FSC	
POINT FORM	(PNTFORM)	<input checked="" type="text"/> FULL	COMPONNT	
POINT DESCRIPTOR	(PTDESC)			
E.U. DESCRIPTOR	(EUDESC)			
POINT KEYWORD	(KEYWORD)			
ASSOCIATED DISPLAY	(ASSOCDSP)			
UNIT ID	(UNIT)			
NETWORK NUMBER	(NTWKNUM)			
F1=PED F3= F5=OVERWRITE F7=RECON F9 =WLK BACK F11=TAB F2=RECALL DISP F4= F6= F8=PED STATUS F10=WRITE F12=LOAD				

14582-A

Page 2 of 5

26 Jul 99 15:47:49 4

PED >>>>>> POINT:		UNIT:	ERRS UNENTERED	PAGE 02 OF 05
NODE NUMBER	(NODENUM)			
SLOT NUMBER	(SLOTNUM)			
PRIMARY MODULE POINT ID	(PRIMMOD)	-----		
F1=PED F3= F5=OVERWRITE F7=RECON F9 =WLK BACK F11=TAB F2=RECALL DISP F4= F6= F8=PED STATUS F10=WRITE F12=LOAD				

14583-A

Continued on next page

A.3 Analog Output, Continued

Page 3 of 5

26 Jul 99 15:48:29 4

PED >>>>>> POINT:		UNIT:	ERRS	PAGE 03 OF 05	
SM DESTINATION ADDRESS	(PLCADDR)	0			
F1=PED	F3=	F5=OVERWRITE	F7=RECON	F9 =WLK BACK	F11=TAB
F2=RECALL DISP	F4=	F6=	F8=PED STATUS	F10=WRITE	F12=LOAD

14214-A

Page 4 of 5

26 Jul 99 15:51:12 4

PED >>>>>> POINT:		UNIT:	ERRS	PAGE 04 OF 05	
NIM-OPERATING CONFIGURATION					
ANALOG OUTPUT					
OUTPUT RAW HIGH LIMIT	(OPFINHI)	4095.0			
OUTPUT RAW LOW LIMIT	(OPFINLO)	0.0			
OUTPUT DIRECTION	(OPTDIR)	☒ DIRECT ☐ REVERSE			
OP TOLERANCE IN %	(\$OPTOL)	0.0			
F1=PED	F3=	F5=OVERWRITE	F7=RECON	F9 =WLK BACK	F11=TAB
F2=RECALL DISP	F4=	F6=	F8=PED STATUS	F10=WRITE	F12=LOAD

14215-A

Continued on next page

A.3 Analog Output, Continued

Page 5 of 5

26 Jul 99 15:52:07 4

PED >>>>>> POINT:	UNIT:	ERRS	PAGE 05 OF 05
NIM-MODE CONFIGURATION			
ANALOG OUTPUT			
REMOTE CASCADE OPTION	(RCASOPT)	NONE	DDC
NORMAL MODE	(NMODE)	NONE	MAN
MODE PERMISSIVE	(MODEPERM)	PERMIT	NOTPERM
F1=PED F3= F5=OVERWRITE F7=RECON F9 =WLK BACK F11=TAB			
F2=RECALL DISP F4= F6= F8=PED STATUS F10=WRITE F12=LOAD			

17804

A.4 Digital Input, Continued

Page 3 of 6

29 Sep 99 09:53:07 5

PED >>>>>> POINT:		UNIT:	ERRS UNENTERED	PAGE 03 OF 06
SM SOURCE ADDRESS	(PLCADDR)	0		
F1=PED	F3=	F5=OVERWRITE	F7=RECON	F9 =WLK BACK
F2=RECALL DISP	F4=	F6=	F8=PED STATUS	F10=WRITE
F11=TAB	F12=LOAD			

14196-A

Page 4 of 6

29 Sep 99 09:54:02 5

PED >>>>>> POINT:		UNIT:	ERRS	PAGE 04 OF 06
NIM-OPERATING CONFIGURATION				
DIGITAL INPUT				
STATE 1 DESCRIPTOR	(STATETXT(1))	ON		
STATE 0 DESCRIPTOR	(STATETXT(0))	OFF		
STATE 1 BOX COLOR	(BOXCLR(1))	RED	GREEN	WHITE
		CYAN	YELLOW	BLUE
STATE 0 BOX COLOR	(BOXCLR(0))	RED	GREEN	WHITE
		CYAN	YELLOW	BLUE
INPUT DIRECTION	(INPTDIR)	DIRECT	REVERSE	
PV SOURCE OPTION	(PVSRCOPT)	ONLYAUTO	ALL	
F1=PED	F3=	F5=OVERWRITE	F7=RECON	F9 =WLK BACK
F2=RECALL DISP	F4=	F6=	F8=PED STATUS	F10=WRITE
F11=TAB	F12=LOAD			

17805

Continued on next page

A.4 Digital Input, Continued

Page 5 of 6

29 Sep 99 09:55:18 5

PED >>>>>> POINT:	UNIT:	ERRS	PAGE 05 OF 06
-------------------	-------	------	---------------

NIM-ALARMING DISPLAY
DIGITAL INPUT

AUXILIARY UNIT (\$AUXUNIT) --

ALARMING OPTION (ALMOPT) NONE OFFNORML CHNGOFST

EVENT REPORT OPTION (EVTOPT) NONE EIP SOE EIP-SOE

F1=PED F3= F5=OVERWRITE F7=RECON F9 =WLK BACK F11=TAB
F2=RECALL DISP F4= F6= F8=PED STATUS F10=WRITE F12=LOAD

17806

Page 6 of 6

29 Sep 99 09:56:14 5

PED >>>>>> POINT:	UNIT:	ERRS	PAGE 06 OF 06
-------------------	-------	------	---------------

CONTACT CUTOUT SOURCE (CCSRC) 0

F1=PED F3= F5=OVERWRITE F7=RECON F9 =WLK BACK F11=TAB
F2=RECALL DISP F4= F6= F8=PED STATUS F10=WRITE F12=LOAD

14197-A

Continued on next page

A.4 Digital Input, Continued

ATTENTION SOE must be chosen for points to be collected by TRICON DISOE. This is how DISOE points are configured. This differs from other TRICON SOE blocks (1-14).

The SOE block number is transparent at this point and data points fall into an SOE block based on which block number (15 or 16) was configured for this SM's UCN node address.

A.5 Digital Output, Continued

Page 3 of 5

29 Sep 99 10:01:08 5

PED >>>>> POINT:		UNIT:	ERRS UNENTERED	PAGE 03 OF 05
SM DESTINATION ADDRESS	(PLCADDR)	0		
DIGITAL OUTPUT TYPE	(DOTYPE)	STATUS		
F1=PED	F3=	F5=OVERWRITE	F7=RECON	F9 =WLK BACK
F2=RECALL DISP	F4=	F6=	F8=PED STATUS	F10=WRITE
				F11=TAB
				F12=LOAD

17807

Page 4 of 5

29 Sep 99 10:01:08 5

PED >>>>> POINT:		UNIT:	ERRS UNENTERED	PAGE 03 OF 05
SM DESTINATION ADDRESS	(PLCADDR)	0		
DIGITAL OUTPUT TYPE	(DOTYPE)	STATUS		
F1=PED	F3=	F5=OVERWRITE	F7=RECON	F9 =WLK BACK
F2=RECALL DISP	F4=	F6=	F8=PED STATUS	F10=WRITE
				F11=TAB
				F12=LOAD

17808

Continued on next page

A.5 Digital Output, Continued

Page 5 of 5

29 Sep 99 10:05:47 5

```

PED >>>>>> POINT: UNIT: ERRS UNENTERED PAGE 05 OF 05
NIM-MODE CONFIGURATION
DIGITAL OUTPUT

NORMAL MODE ATTRIBUTE (NMODATTR) NONE OPERATOR PROGRAM
MODE PERMISSIVE (MODEPERM) PERMIT NOTPERM

F1=PED F3= F5=OVERWRITE F7=RECON F9 =WLK BACK F11=TAB
F2=RECALL DISP F4= F6= F8=PED STATUS F10=WRITE F12=LOAD

```

A.6 Digital Composite

Page 1 of 9

29 Sep 99 10:11:08 5

PED >>>>>> POINT:		UNIT:	ERRS UNENTERED	PAGE 01 OF 09
NIM-POINT ASSIGNMENT DIGITAL COMPOSITE				
TAG NAME	(NAME)			
NODE TYPE	(NODETYP)	PM	APM	HPM
		LM	SM	
SM PLATFORM	(SMPLTFM)	TRICON	FSC	
POINT FORM	(PNTFORM)	FULL	COMPONNT	
POINT DESCRIPTOR	(PTDESC)			
E.U. DESCRIPTOR	(EUDESC)			
POINT KEYWORD	(KEYWORD)			
ASSOCIATED DISPLAY	(ASSOCDSP)			
UNIT ID	(UNIT)	☐		
NETWORK NUMBER	(NTWKNUM)	☐		
F1=PED F3= F5=OVERWRITE F7=RECON F9 =WLK BACK F11=TAB F2=RECALL DISP F4= F6= F8=PED STATUS F10=WRITE F12=LOAD				

14588-A

Page 2 of 9

29 Sep 99 10:12:13 5

PED >>>>>> POINT:		UNIT:	ERRS	PAGE 02 OF 09
NODE NUMBER	(NODENUM)	☐		
SLOT NUMBER	(SLOTNUM)			
PRIMARY MODULE POINT ID	(PRIMMOD)	-----		
NUMBER OF STATES	(NOSTATES)	2	3	
NUMBER OF INPUTS	(NODINPTS)	0	1	2
NUMBER OF OUTPUTS	(NODOPTS)	0	1	2 3
F1=PED F3= F5=OVERWRITE F7=RECON F9 =WLK BACK F11=TAB F2=RECALL DISP F4= F6= F8=PED STATUS F10=WRITE F12=LOAD				

14589-A

Continued on next page

A.6 Digital Composite, Continued

Page 3 of 9

29 Sep 99 10:26:08 5

PED >>>>>> POINT:		UNIT:	ERRS	PAGE 03 OF 09	
NIM-STATE CONFIGURATION					
DIGITAL COMPOSITE					
STATE 1 DESCRIPTOR	(STATETXT(1))	ON			
STATE 0 DESCRIPTOR	(STATETXT(0))	OFF			
STATE 1 BOX COLOR	(BOXCLR(1))	RED	<input checked="" type="text" value="GREEN"/>	WHITE	BLACK
		CYAN	YELLOW	BLUE	MAGENTA
STATE 0 BOX COLOR	(BOXCLR(0))	RED	GREEN	WHITE	BLACK
		CYAN	<input checked="" type="text" value="YELLOW"/>	BLUE	MAGENTA
MOMENTARY STATES	(MOMSTATE)	<input checked="" type="text" value="NONE"/> MOM_0 MOM_1			
LOGIC SOURCE	(LOGICSRC)	-----			
F1=PED F3= F5=OVERWRITE F7=RECON F9 =WLK BACK F11=TAB					
F2=RECALL DISP F4= F6= F8=PED STATUS F10=WRITE F12=LOAD					

14202-A

Page 4 of 9

29 Sep 99 10:27:03 5

PED >>>>>> POINT:		UNIT:	ERRS	PAGE 04 OF 09	
NIM-PV/INPUT CONFIGURATION					
DIGITAL COMPOSITE					
INPUT CONNECTIONS					

	SOURCE				
	POINT.PARAMETER				
	DISRC(N)				
INPUT					DESTINATION
1					D1
INPUT STATE CONFIGURATION					

D1=ON	(D1_1)	<input checked="" type="text" value="PVSTATE1"/>	PVSTATE0		
PV SOURCE OPTION	(PVSRCOPT)	<input checked="" type="text" value="ONLYAUTO"/> ALL			
F1=PED F3= F5=OVERWRITE F7=RECON F9 =WLK BACK F11=TAB					
F2=RECALL DISP F4= F6= F8=PED STATUS F10=WRITE F12=LOAD					

14203-A

Continued on next page

A.6 Digital Composite, Continued

Page 5 of 9

29 Sep 99 10:27:51 5

PED >>>>>> POINT:		UNIT:	ERRS	PAGE 05 OF 09	
NIM-OUTPUT CONFIGURATION					
DIGITAL COMPOSITE					
OUTPUT STATE CONFIGURATION					

OUTPUT1					

STATE1	☐ ON	☐ OFF			
	(ST1_OP1)				
STATE0	☐ ON	☐ OFF			
	(ST0_OP1)				
OUTPUT CONNECTIONS					

OUTPUT	DESTINATION POINT_ID.PARAMETER				
	DODSTN(N)				
1					
F1=PED F3= F5=OVERWRITE F7=RECON F9 =WLK BACK F11=TAB					
F2=RECALL DISP F4= F6= F8=PED STATUS F10=WRITE F12=LOAD					

14204-A

Page 6 of 9

29 Sep 99 10:28:38 5

PED >>>>>> POINT:		UNIT:	ERRS	PAGE 06 OF 09	
INTERLOCKS					

	PERMISSIVES		OVERRIDES		
	PISRC(N)		OISRC(N)		
STATE1	P1	☐	I1	☐	
STATE0	P0	☐	I0	☐	
LOCAL MANUAL SOURCE	(LMSRC)	☐			
F1=PED F3= F5=OVERWRITE F7=RECON F9 =WLK BACK F11=TAB					
F2=RECALL DISP F4= F6= F8=PED STATUS F10=WRITE F12=LOAD					

14205-A

Continued on next page

A.6 Digital Composite, Continued

Page 7 of 9

```

29 Sep 99 10:35:00 5
PED >>>>>> POINT: UNIT: ERRS PAGE 07 OF 09
NIM-MODE CONFIGURATION
DIGITAL COMPOSITE

NORMAL MODE ATTRIBUTE (NMODATTR) NONE OPERATOR PROGRAM
MODE PERMISSIVE (MODEPERM) PERMIT NOTPERM
INTERLOCK BYPASS OPTION (OROPT) OFF ON

F1=PED F3= F5=OVERWRITE F7=RECON F9 =WLK BACK F11=TAB
F2=RECALL DISP F4= F6= F8=PED STATUS F10=WRITE F12=LOAD
14206-

```

Page 8 of 9

```

29 Sep 99 10:36:03 5
PED >>>>>> POINT: UNIT: ERRS PAGE 08 OF 09
NIM-ALARMING DISPLAY
DIGITAL COMPOSITE

AUXILIARY UNIT ($AUXUNIT) --

ALARMING OPTION (ALMOPT) NONE CMDDIS OFFNORML

BAD PV ALARM PRIORITY (BADPVPR) NOACTION JOURNAL LOW HIGH

EMERGENCY PRINTER JNLPRINT

BAD CONTROL ALARM PRIORITY (BADCTLPR) NOACTION JOURNAL LOW HIGH

EMERGENCY PRINTER JNLPRINT

EVENT REPORT OPTION (EVTOPT) NONE EIP

F1=PED F3= F5=OVERWRITE F7=RECON F9 =WLK BACK F11=TAB
F2=RECALL DISP F4= F6= F8=PED STATUS F10=WRITE F12=LOAD
14207-

```

Continued on next page

A.6 Digital Composite, Continued

Page 9 of 9

29 Sep 99 10:37:06 5

PED >>>>>>	POINT:	UNIT:	ERRS	PAGE 09 OF 09
CONTACT CUTOUT SOURCE (CCSRC) 0				
F1=PED F2=RECALL DISP F3= F4= F5=OVERWRITE F6= F7=RECON F8=PED STATUS F9 =WLK BACK F10=WRITE F11=TAB F12=LOAD				

17810

A.7 Logic

Page 1 of 7

29 Sep 99 10:38:12 5

PED >>>>>> POINT:UNIT: ERRS UNENTERED PAGE 01 OF 07

NIM-POINT ASSIGNMENT

LOGIC

TAG NAME

(NAME)

NODE TYPE

(NODETYP)

PM

APM

HPM

LM

SM

SM PLATFORM

(SMPLTFM)

TRICON

FSC

POINT FORM

(PNTFORM)

FULL

COMPONNT

POINT DESCRIPTOR

(PTDESC)

ASSOCIATED DISPLAY

(ASSOCDSP)

UNIT ID

(UNIT)

NETWORK NUMBER

(NTWKNUM)

NODE NUMBER

(NODENUM)

SLOT NUMBER

(SLOTNUM)

F1=PED

F3=

F5=OVERWRITE

F7=RECON

F9 =WLK BACK

F11=TAB

F2=RECALL DISP

F4=

F6=

F8=PED STATUS

F10=WRITE

F12=LOAD

14590-A

Page 2 of 7

29 Sep 99 10:39:03 5

PED >>>>>> POINT:UNIT: ERRS PAGE 02 OF 07

PRIMARY MODULE POINT ID

(PRIMMOD)

LOGIC MIX

(LOGMIX)

12_0_12

F1=PED

F3=

F5=OVERWRITE

F7=RECON

F9 =WLK BACK

F11=TAB

F2=RECALL DISP

F4=

F6=

F8=PED STATUS

F10=WRITE

F12=LOAD

14591-A

Continued on next page

A.7 Logic, Continued

Page 3 of 7

29 Sep 99 10:39:42 5

PED >>>>>> POINT:	UNIT:	ERRS	PAGE 03 OF 07
NIM-INPUT CONNECTIONS			
LOGIC			
INPUT #1 CONNECTION	(LISRC(1))		
INPUT #2 CONNECTION	(LISRC(2))		
INPUT #3 CONNECTION	(LISRC(3))		
INPUT #4 CONNECTION	(LISRC(4))		
INPUT #5 CONNECTION	(LISRC(5))		
INPUT #6 CONNECTION	(LISRC(6))		
INPUT #7 CONNECTION	(LISRC(7))		
INPUT #8 CONNECTION	(LISRC(8))		
INPUT #9 CONNECTION	(LISRC(9))		
INPUT #10 CONNECTION	(LISRC(10))		
INPUT #11 CONNECTION	(LISRC(11))		
F1=PED F3= F5=OVERWRITE F7=RECON F9 =WLK BACK F11=TAB F2=RECALL DISP F4= F6= F8=PED STATUS F10=WRITE F12=LOAD			

Page 4 of 7

```

29 Sep 99 10:41:07 5
PED >>>>> POINT: UNIT: ERRS PAGE 04 OF 07
INPUT #12 CONNECTION (LISRC(12)) 
LOGICAL INPUT (LIBADOPT) OFF ON HOLD
FAILURE HANDLING OPTION

F1=PED F3= F5=OVERWRITE F7=RECON F9 =WLK BACK F11=TAB
F2=RECALL DISP F4= F6= F8=PED STATUS F10=WRITE F12=LOAD
14219-A

```

Continued on next page

A.7 Logic, Continued

Page 5 of 7

29 Sep 99 10:41:51 5

PED >>>>> POINT:		UNIT: ERRS	PAGE 05 OF 07
NIM-LOGIC OUTPUT CONNECTIONS			
LOGIC			
	DESTINATION POINT_ID.PARAMETER LODSTN(N)	ENABLE LOENBL(N)	
1			
2			
3			
4			
5			
6			
7			
8			
9			
F1=PED F3= F5=OVERWRITE F7=RECON F9 =WLK BACK F11=TAB F2=RECALL DISP F4= F6= F8=PED STATUS F10=WRITE F12=LOAD			

14220-A

Page 6 of 7

29 Sep 99 10:42:40 5

PED >>>>> POINT:		UNIT: ERRS UNENTERED	PAGE 06 OF 07
10			
11			
12			
F1=PED F3= F5=OVERWRITE F7=RECON F9 =WLK BACK F11=TAB F2=RECALL DISP F4= F6= F8=PED STATUS F10=WRITE F12=LOAD			

17811

Continued on next page

A.7 Logic, Continued

Page 7 of 7

29 Sep 99 10:42:40 5

PED >>>>>>	POINT:	UNIT:	ERRS UNENTERED	PAGE 06 OF 07
10				0
11				0
12				0
F1=PED F3= F5=OVERWRITE F7=RECON F9 =WLK BACK F11=TAB F2=RECALL DISP F4= F6= F8=PED STATUS F10=WRITE F12=LOAD				

17812

A.8 Flag, Continued

Page 3 of 3

29 Sep 99 10:46:42 5

PED >>>>> POINT:		UNIT:	ERRS	PAGE 03 OF 03	
NIM-OPERATING CONFIGURATION					
FLAG					
STATE 1 DESCRIPTOR	(STATETXT(1))	ON			
STATE 0 DESCRIPTOR	(STATETXT(0))	OFF			
STATE 1 BOX COLOR	(BOXCLR(1))	RED	GREEN	WHITE	BLACK
		CYAN	YELLOW	BLUE	MAGENTA
STATE 0 BOX COLOR	(BOXCLR(0))	RED	GREEN	WHITE	BLACK
		CYAN	YELLOW	BLUE	MAGENTA
F1=PED F3= F5=OVERWRITE F7=RECON F9 =WLK BACK F11=TAB					
F2=RECALL DISP F4= F6= F8=PED STATUS F10=WRITE F12=LOAD					

17813

A.9 Timer

Page 1 of 4

```

29 Sep 99 10:48:22 5
PED >>>>>> POINT: UNIT: ERRS UNENTERED PAGE 01 OF 04
NIM-POINT ASSIGNMENT
  TIMER

TAG NAME (NAME) 

MODE TYPE (NODETYP)  PM  APM  HPM
                   LM  SM

SM PLATFORM (SMPLTFM)  TRICON  FSC

POINT FORM (PNTFORM)  FULL  COMPONNT

POINT DESCRIPTOR (PTDESC) 

E.U. DESCRIPTOR (EUDESC) 

POINT KEYWORD (KEYWORD) 

ASSOCIATED DISPLAY (ASSOCDSP) 

UNIT ID (UNIT) ☐

NETWORK NUMBER (NTWKNUM) ☐

F1=PED F3= F5=OVERWRITE F7=RECON F9 =WLK BACK F11=TAB
F2=RECALL DISP F4= F6= F8=PED STATUS F10=WRITE F12=LOAD
14595-

```

Page 2 of 4

```

29 Sep 99 10:49:10 5
PED >>>>> POINT: UNIT: ERRS PAGE 02 OF 04
NODE NUMBER (NODENUM) 
SLOT NUMBER (SLOTNUM) 
PRIMARY MODULE POINT ID (PRIMMOD) 

F1=PED F3= F5=OVERWRITE F7=RECON F9 =WLK BACK F11=TAB
F2=RECALL DISP F4= F6= F8=PED STATUS F10=WRITE F12=LOAD
14596-A

```

Continued on next page

A.9 Timer, Continued

Page 3 of 4

29 Sep 99 10:49:45 5

PED >>>>>> POINT:	UNIT:	ERRS	PAGE 03 OF 04
SM TARGET ADDRESS	(PLCADDR)	0	
SM ACCUMULATOR ADDRESS	(PVSLT SRC)	0	
SM SETPOINT ADDRESS	(SPSLT SRC)	0	
F1=PED	F3=	F5=OVERWRITE	F7=RECON
F2=RECALL DISP	F4=	F6=	F8=PED STATUS
			F9 =WLK BACK
			F10=WRITE
			F11=TAB
			F12=LOAD

14225-A

Page 4 of 4

29 Sep 99 10:50:40 5

PED >>>>>> POINT:	UNIT:	ERRS	PAGE 04 OF 04
NIM-OPERATING CONFIGURATION			
TIMER			
TIMER BASE PERIOD	(TIMEBASE)	SECONDS	MINUTES
PV DECIMAL FORMAT	(PVFORMAT)	00	01 02
F1=PED	F3=	F5=OVERWRITE	F7=RECON
F2=RECALL DISP	F4=	F6=	F8=PED STATUS
			F9 =WLK BACK
			F10=WRITE
			F11=TAB
			F12=LOAD

17814

A.10 Numeric, Continued

Page 3 of 3

29 Sep 99 10:54:46 5

PED >>>>> POINT:		UNIT:	ERRS	PAGE 03 OF 03	
NIM-PV CONFIGURATION					
NUMERIC					
PV DECIMAL FORMAT		(PVFORMAT)	00	<input checked="" type="text" value="01"/>	02
F1=PED	F3=	F5=OVERWRITE	F7=RECON	F9 =WLK BACK	F11=TAB
F2=RECALL DISP	F4=	F6=	F8=PED STATUS	F10=WRITE	F12=LOAD

14228-A

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